

# **1 Irrigation strategies under contrasting water-storage and rainfall**

## **2 conditions in the North China Plain**

3 Linjia Yao<sup>1,†</sup>, Bin Chen<sup>2,†</sup>, Hao Feng<sup>2</sup>, Qiang Yu<sup>2,\*</sup>, Gang Zhao<sup>2,\*</sup>

4 <sup>1</sup> College of Natural Resources and Environment, Northwest A&F University, Yangling 712100, China

5 <sup>2</sup> State Key Laboratory of Soil and Water Conservation and Desertification Control, College of Soil and Water  
6 Conservation Science and Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China

7 † Linjia Yao and Bin Chen contributed equally to this work and share first authorship.

8 \* Corresponding authors: Qiang Yu (yuq@nwafu.edu.cn) and Gang Zhao (gang.zhao@nwafu.edu.cn).

### **9 Cover letter**

10 Dear Editor,

11 Please consider the research article “Irrigation strategies under contrasting water-storage and rainfall conditions in  
12 the North China Plain” for publication in Agricultural Water Management.

13 Reducing irrigation while sustaining grain production requires balancing water consumption and the distribution of  
14 production losses. Using a process-based crop model, the study compares uniform and spatially targeted irrigation  
15 under equal water budgets and tests whether satellite water-storage information improves rainfall-based irrigation  
16 decisions.

17 At the same 50% irrigation reduction, targeting retained 4.87% more simulated grain than uniform cuts, but  
18 increased evapotranspiration and reduced grain production in some locations. Adding regional storage information  
19 did not change irrigation quotas or outcomes relative to matched rainfall-only strategies within the tested design.

20 These model-based findings support irrigation strategies that combine production goals with limits on consumptive  
21 use and safeguards for local production. Their focus on irrigation allocation and crop water use makes the study  
22 well suited to Agricultural Water Management.

23 Thank you for considering this manuscript.

24 Sincerely,

25 Qiang Yu and Gang Zhao

26 State Key Laboratory of Soil and Water Conservation and Desertification Control, College of Soil and Water  
27 Conservation Science and Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China  
28 yuq@nwafu.edu.cn; gang.zhao@nwafu.edu.cn

## 29 **Highlights**

- 30 • Irrigation cuts require balancing grain retention, water use and local losses.
- 31 • Targeting added 0.732 t ha<sup>-1</sup> dry grain and 20.76 mm ET yearly over uniform 50% cuts.
- 32 • At 50% cuts, mean grain was below uniform allocation on 37.60% of the area.
- 33 • Over nine GRACE years, grain retention was 96.27% (95% target) and 99.63% (98%).
- 34 • Matched rainfall-only controls differed in grain production by 0% at both targets.

## 35 **Manuscript**

### 36 **Abstract**

37 Efficient irrigation allocation is essential for sustaining grain production while limiting water consumption in the  
38 North China Plain wheat–maize rotation. How spatial targeting redistributes production losses, and whether  
39 regional water-storage information improves rainfall-based irrigation decisions, remain unresolved. Continuous  
40 process-based crop simulations compared uniform and targeted allocations under equal irrigation budgets and  
41 evaluated annual storage–rainfall strategies against matched rainfall-only controls during 2014–2025. Growth  
42 parameters were fitted to multisite observations; water-use parameters were calibrated against Wuqiao treatments in  
43 2016–2018 and retrospectively tested in 2019. At a 50% irrigation reduction, targeting retained 0.73 t ha<sup>-1</sup> yr<sup>-1</sup>  
44 more simulated combined dry grain than uniform cuts across 9.87 million rotation hectares, equivalent to 7.22  
45 million t yr<sup>-1</sup> or 4.87% of uniform production. Mean production was lower than under uniform cuts on 37.60% of  
46 mapped area, while crop-plus-fallow evapotranspiration increased by 20.76 mm yr<sup>-1</sup> (2.05 km<sup>3</sup> yr<sup>-1</sup>). Across nine  
47 years with eligible Gravity Recovery and Climate Experiment (GRACE) observations, strategies selected at 95%  
48 and 98% grain targets retained 96.27% and 99.63% of conventional production, respectively. The 95% strategy  
49 reduced irrigation by 130.72 mm and evapotranspiration by 36.84 mm per rotation. Matched storage–rainfall and

50 rainfall-only strategies selected identical quotas and produced identical outcomes. These conditional results support  
51 joint production and consumptive-use constraints and local production safeguards, while showing that additional  
52 storage information provides no decision benefit within the tested quota design.

## 53 **Keywords**

54 winter wheat–summer maize rotation; irrigation management; crop modeling; GRACE; water availability;  
55 consumptive water use

## 56 **1. Introduction**

57 Irrigation management in the North China Plain must reconcile grain production with water-resource constraints  
58 across the winter wheat–summer maize rotation (Yang et al., 2022). Wheat grows through a relatively dry season,  
59 whereas maize receives more summer rainfall; soil-water reserves connect their responses to irrigation (Yang et al.,  
60 2024). Long-term well-watered observations show that rising grain yield has accompanied increasing actual  
61 evapotranspiration (ET), despite comparatively stable reference ET (Zhang et al., 2011). Maintaining production  
62 with less irrigation therefore requires attention to both the water delivered to fields and the water consumed over the  
63 complete rotation.

64 The timing and location of irrigation supply determine its marginal production return. A sixteen-year experiment in  
65 2003–2018 compared seven wheat–maize irrigation schedules and found diminishing production gains as irrigation  
66 increased (Yang et al., 2022). A four-year Wuqiao experiment also showed that precipitation conditions influenced  
67 the preferred wheat schedule and grain production in the subsequent maize crop (Yang et al., 2024). These  
68 responses reflect the sequence of crop demand, rainfall and soil-water carry-over across both seasons (Yang et al.,  
69 2024; Yang et al., 2022). At regional scale, irrigation requirements vary among locations and hydrological years  
70 (Ren et al., 2022; Wu et al., 2019). Crop-model optimization and irrigation-allocation studies provide approaches  
71 for using this heterogeneity to guide management (Jiang et al., 2016; Xiao et al., 2024). Under a common water  
72 budget, targeting locations with larger marginal grain returns can improve regional production retention, while  
73 changing the distribution of production losses.

74 Assessing these allocation consequences requires consistent crop and water accounting. Field delivery, ET, return  
75 flows and storage changes describe different quantities, so a reduction in irrigation does not directly establish an  
76 equivalent reduction in consumption (Grafton et al., 2018; Perry et al., 2009). Process-based applications of APSIM  
77 and RZ-SHAW in the North China Plain connect crop production with water productivity, field water balance and

78 ET (Chen et al., 2010; Fang et al., 2014). However, transpiration formulation remains a major source of variation in  
79 simulated wheat water use (Cammarano et al., 2016). Phenology, canopy, biomass, grain and ET observations  
80 therefore constrain complementary processes, while documented irrigation treatments assess their combined  
81 response to water supply. Continuous simulation retains the soil-water carry-over needed to compare complete  
82 rotation histories.

83 Spatial allocation determines where irrigation is supplied, while antecedent hydrological conditions can inform the  
84 annual irrigation budget. Recent rainfall describes water input at the crop-system scale, whereas GRACE total water  
85 storage integrates several terrestrial stores over a regional footprint (Save et al., 2016; Scanlon et al., 2016). Recent  
86 aquifer recovery under water transfers, abstraction management and changing climate illustrates the broader context  
87 of regional storage variation (Long et al., 2025; Xu et al., 2026). Storage information may thus characterize  
88 conditions beyond local rainfall. Its practical value for irrigation nevertheless depends on whether it changes the  
89 selected action or improves the resulting grain–water outcome relative to rainfall-based decisions.

90 Two connected management questions remain unresolved. First, it is unclear how much grain spatial targeting can  
91 retain under an identical irrigation budget, and what consumptive-water costs and persistent local production losses  
92 accompany that regional benefit. Rotation-wide accounting must retain soil-water carry-over to assess these  
93 outcomes together. Second, the additional decision value of regional storage information beyond antecedent rainfall  
94 remains uncertain. A storage indicator can distinguish hydrological conditions without changing an irrigation  
95 choice. Comparisons using the same production targets, candidate quotas and monitoring availability are needed to  
96 isolate its contribution to management outcomes.

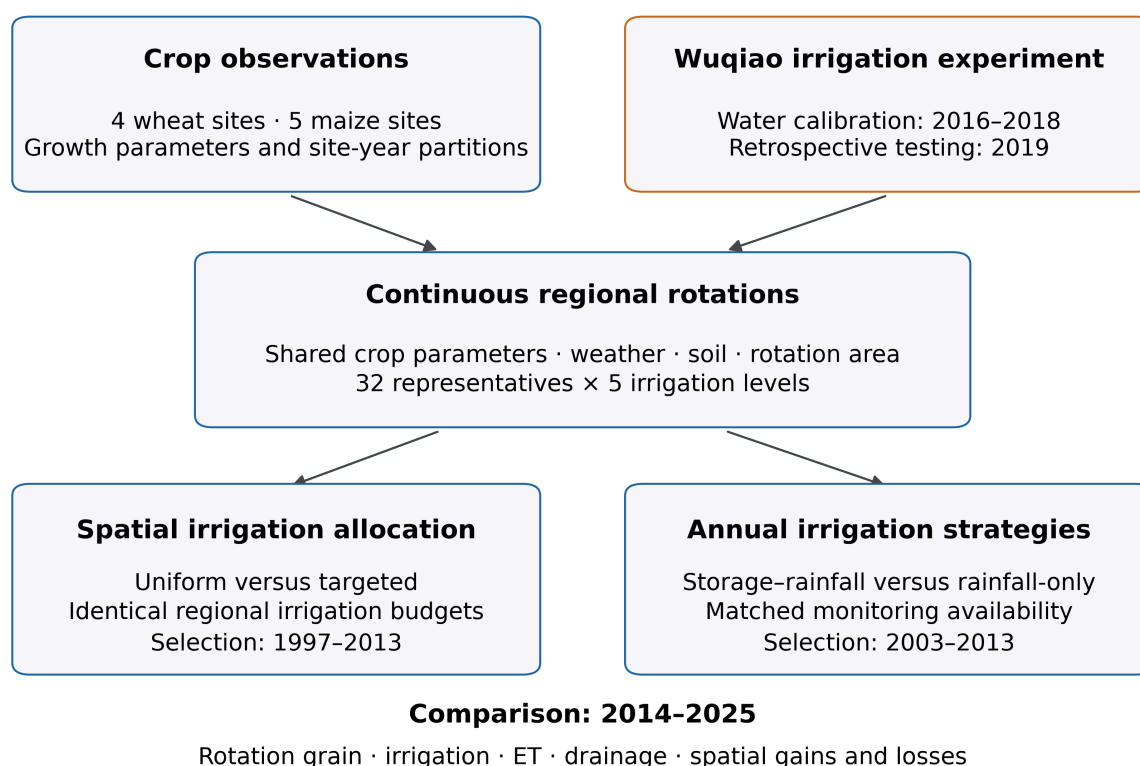
97 The objectives were to (i) quantify rotation grain production, crop-plus-fallow ET and profile drainage under  
98 uniform and spatially targeted irrigation allocations with common regional water budgets; (ii) characterize the  
99 spatial distribution, magnitude and annual persistence of allocation-related grain gains and losses; and (iii) assess  
100 whether antecedent regional storage information changes annual irrigation quotas and grain–water outcomes  
101 relative to rainfall-only controls under matched selection and monitoring conditions. Continuous Open Crop Model  
102 simulations of the North China Plain wheat–maize rotation, parameterized using multisite crop observations and  
103 documented irrigation treatments, provide the basis for these conditional comparisons during 2014–2025.

## 104 **2. Materials and methods**

### 105 **2.1. Overall workflow**

106 The framework connects crop-response estimation with spatial allocation and annual irrigation-quota selection  
 107 (Figure 1). Multisite crop observations support growth-parameter estimates, while documented Wuqiao treatments  
 108 in 2016–2018 constrain eight water coefficients per crop. Wuqiao 2019 provides retrospective testing; station  
 109 observations characterize transfer under the recorded management and initialization. Observation coverage and the  
 110 distinct estimation and evaluation partitions are summarized in Figure 3. Daily weather, mapped rotation area and  
 111 layered soil profiles define 32 regional representatives.

112 The same crop and water parameters drive continuous crop and fallow histories under five irrigation levels. Spatial  
 113 allocations selected from 1997–2013 responses share identical regional delivery budgets with uniform reductions.  
 114 Annual storage–rainfall and rainfall-only strategies are selected from matched 2003–2013 responses and compared  
 115 with common monitoring availability. Grain production, field irrigation, crop-plus-fallow ET and profile drainage  
 116 are evaluated during 2014–2025.

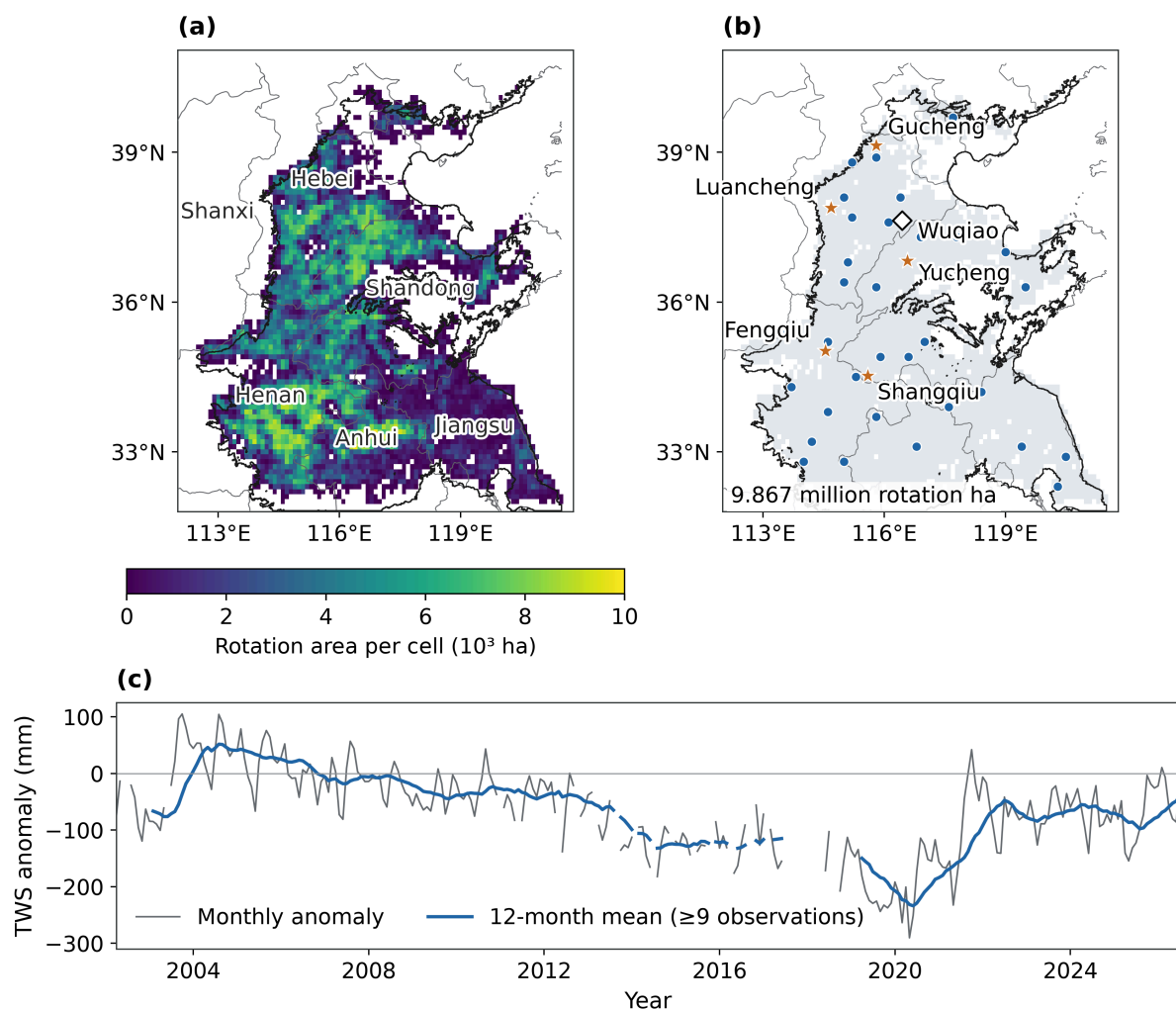


117 **Figure 1. Study workflow.** Multisite observations support growth parameters, while documented Wuqiao treatments calibrate water-  
 118 use parameters and provide retrospective testing. Shared parameters drive continuous regional rotations. Spatial allocation  
 119 compares uniform and targeted irrigation under identical budgets; annual strategies compare storage–rainfall and rainfall-only  
 120 selection under matched monitoring availability. Outcomes cover 2014–2025.

## 121 2.2. Study area

122 The study area covers the winter-wheat–summer-maize lowlands of the North China Plain within 112–121.5°E and  
123 32–41°N, including the Beijing–Tianjin–Hebei sector and the Henan–Anhui–Jiangsu lowlands (Figure 2).  
124 Catchment screening, elevation  $\leq 200$  m and slope  $\leq 2^\circ$  define a connected lowland area of 404,658 km<sup>2</sup>. Terrain was  
125 represented at 0.01° resolution using resampled SRTM elevation and derived slope (NASA JPL, 2013). The  
126 boundary is a reproducible agricultural study-area definition; geographical and terrain-threshold sensitivities are  
127 reported in Supplementary Section S3.

128 The 2020 ChinaCP winter-wheat–summer-maize class was corrected by the corresponding CPM cropland  
129 percentage to reduce full-pixel area inflation (Qiu et al., 2022). The resulting rotation footprint was 9,866,733 ha  
130 before soil exclusions. Three cells with wholly missing soil profiles, totaling 131 ha, were excluded, leaving  
131 9,866,602 ha in 3,641 AgERA5 cells. A fixed 2020 footprint applies to all climate years, so modeled totals describe  
132 climatic and management responses over a common area rather than historical production trends. The 2015–2021  
133 area series, terrain-threshold alternatives and a common-extent comparison with CCD2020 quantify mapping and  
134 boundary sensitivity in the supplementary material.



135 **Figure 2. Study domain, rotation area, observation sites and regional water storage. (a) Mapped 2020 wheat–maize rotation area. (b)**  
 136 **Five crop-observation sites, the Wuqiao irrigation experiment and 32 regional simulation units. (c) Observed CSR monthly**  
 137 **terrestrial-water-storage anomalies and a 12-calendar-month mean requiring at least nine observations. Missing months remain**  
 138 **gaps. Thin grey lines show provincial boundaries; black outlines delimit the operational physiographic study domain. Maps use an**  
 139 **equiarectangular projection (36°N, 117°E). Provincial boundaries: Natural Earth Admin 1 v5.1.1 (Natural Earth, n.d.).**

## 140 **2.3. Field observations and environmental data**

### 141 **2.3.1. Multisite crop observations**

142 Winter-wheat and summer-maize observations were assembled from the National Ecosystem Science Data Center  
 143 and linked archives at Fengqiu, Gucheng, Luancheng, Shangqiu and Yucheng. The sources contain long-term crop  
 144 monitoring and the Gucheng maize sowing-date experiment. Archived crop-season groups span 2004–2008 at  
 145 Fengqiu and Luancheng, 2018–2020 at Gucheng, 2001–2007 at Shangqiu and 1998–2022 at Yucheng. Coverage is  
 146 intermittent; exact crop-by-site years and dataset identifiers are listed in Tables S1 and S2.

147 The archive retains 132 crop–plot–season trajectories in 71 original calibration and 61 retrospective-evaluation  
148 cases (Figure 3). Quality-controlled targets occur in 127 cases. The 7,397 scored records comprise 795 LAI  
149 measurements, 543 in-season aboveground-biomass measurements, 5,675 daily ET measurements, 192 grain-yield  
150 samples and 192 harvest-biomass samples (Table 1). Daily ET is available at Fengqiu and Yucheng; grain and  
151 harvest-biomass samples are confined to Yucheng. Six quadrats within a crop-year characterize sampling variation  
152 rather than six independent growing seasons.

153 Observations and predictions share recorded sampling dates or documented averaging windows. Stage-indexed  
154 Gucheng LAI measurements use the corresponding observed stage dates as sampling coordinates; those dates do  
155 not prescribe simulated development. Fengqiu ET identifies the lysimeter observation plot. Yucheng ET retains the  
156 measurement definitions of its source archive. Provider grain and aboveground dry masses in  $\text{g m}^{-2}$  are multiplied  
157 by 10 to obtain  $\text{kg ha}^{-1}$ .

158 Eligibility requires valid dates, compatible crop and plot identities, consistent stage sequences and documented  
159 quantity units. Wheat LAI above  $7 \text{ m}^2 \text{ m}^{-2}$  is excluded under the study-specific screening rule, with raw values and  
160 reasons preserved. The six-replicate maize harvest vectors in Yucheng 2012 and 2017 are identical despite different  
161 dates and cultivar labels. Both groups remain excluded because their source authenticity is unresolved; 12 samples  
162 account for 24 grain and harvest-biomass records. Other observations and all eligible sites remain available.

163 Complete irrigation histories matched to station plots remain unconfirmed. Station growth and ET consequently  
164 supply conditional transfer comparisons; no station observation enters the current water-parameter objective.  
165 Twenty-six Fengqiu cases retain complete 0–1.5 m moisture profiles from the preceding calendar month: 11 in the  
166 original calibration partition and 15 in retrospective evaluation. Volumetric percentages from fifteen 100-mm source  
167 layers are conservatively remapped to the model layers without depth extrapolation or clipping. Their monthly  
168 timing and subplot support do not establish a measured sowing-day state. Cases without an eligible profile retain  
169 their archived initialization.

170 **Table 1. Eligible station-observation records by site and variable.**

| Site      | LAI | Biomass | Daily ET | Yield | Harvest biomass |
|-----------|-----|---------|----------|-------|-----------------|
| Fengqiu   | 159 | 163     | 962      | 0     | 0               |
| Gucheng   | 216 | 0       | 0        | 0     | 0               |
| Luancheng | 123 | 165     | 0        | 0     | 0               |
| Shangqiu  | 114 | 148     | 0        | 0     | 0               |
| Yucheng   | 183 | 67      | 4713     | 192   | 192             |

171 Counts denote individual observations, not independent experiments. Station comparisons retain inherited growth-  
172 fitting history and unconfirmed complete management. Yucheng grain and harvest biomass contain six dry-matter  
173 samples per crop-year; the duplicated 2012 and 2017 maize groups remain excluded.

174 **2.3.2. Field irrigation experiment**

175 The Wujiao experiment at 37.630°N, 116.444°E contains four wheat irrigation schedules, W0–W3, during the  
176 2015–2019 rotation (Yang et al., 2024). Dated 75-mm events, sowing and harvest dates, nitrogen input of 158.4 kg  
177 ha<sup>-1</sup> per crop and treatment-specific initial total water storage in 2 m are retained. Maize irrigation is identical  
178 among treatments within each year: zero in 2016 and 2018, and 75 mm on 18 June in 2017 and 2019. Its W0–W3  
179 labels therefore identify preceding wheat management. Seasonal initial water distribution among layers is assumed.  
180 Field capacity, wilting point and bulk density are transferred from the published Luancheng profile, with its bottom  
181 10 cm extrapolated (Sun et al., 2006). Conductivity follows a separate SoilGrids–Saxton–Rawls prior (ISRIC —  
182 World Soil Information, 2020; Saxton and Rawls, 2006); the particle-density prior is 2.75 g cm<sup>-3</sup>.

183 Published seasonal ET equals rainfall plus irrigation plus depletion of 0–2 m storage, with runoff and drainage  
184 treated as negligible (Yang et al., 2024). The model represents both outflows. The 2016 maize balance uses 499.1  
185 mm of maize-season rainfall; the 599.6-mm rotation total also contains 100.5 mm during wheat. This period  
186 reconciliation reproduces the four printed maize ET values within 0.9 mm of rounding, and all 32 crop–treatment–  
187 season ET targets remain eligible. Simulations retain their exact archived weather forcing. Wheat crop-start storage  
188 follows the pre-sowing simulation, whereas maize is independently initialized from observed treatment-specific  
189 total storage. Reported wheat-straw mulch has no measured cover fraction, so the maize simulations retain the  
190 exposed-soil evaporation boundary (Yang et al., 2024).

191 Harvest aboveground biomass includes grain and residues cut at ground level and oven-dried to constant mass. Its  
192 dry-matter basis requires no moisture adjustment. Published grain yields are standardized to 13% moisture (Yang et  
193 al., 2024). Annual 2016–2019 means and standard errors are digitized from the source Supplementary Figure S2  
194 and reconciled with its eight printed four-year means. Pixel tolerances are approximately  $0.043 \text{ t ha}^{-1}$  for wheat and  
195  $0.056 \text{ t ha}^{-1}$  for maize. Observed grain is multiplied by 0.87 for dry-matter fitting; model dry grain is divided by  
196 0.87 for comparisons reported at 13% moisture. Annual and four-year means overlap and do not provide separate  
197 independent observations.

198 The 2016–2018 harvest years contribute 12 treatment seasons per crop to the water fit, with seasonal ET, dry grain,  
199 dry aboveground biomass and annual W3–W0 ET contrasts as targets. The four 2019 treatment means per crop are  
200 excluded from fitting and parameter selection. Their prior inspection during model development makes this test  
201 retrospective (Figure 3).

### 202 **2.3.3. Meteorological, soil and spatial data**

203 Daily AgERA5 v2.0 weather data for 1996–2025 supplied minimum and maximum air temperature,  
204 precipitation, solar radiation, reference evapotranspiration, vapor pressure and 10-m wind speed  
205 (Copernicus Climate Change Service, 2020). Reference evapotranspiration ( $ET_0$ ) entered the water-  
206 balance module directly.

207 Native 250 m SoilGrids v2.0 maps of texture, bulk density, organic carbon and coarse fragments were averaged to  
208  $0.01^\circ$ , then aggregated over mapped rotation pixels within  $0.1^\circ$  meteorological cells, for six depth intervals between  
209 0 and 2 m (ISRIC — World Soil Information, 2020). Hydraulic properties were derived using Saxton–Rawls  
210 equations with bulk-density and gravel adjustments (Saxton and Rawls, 2006). Organic matter was approximated  
211 using an assumed organic-matter/organic-carbon ratio of 1.724, and air-dry water content was assumed to equal half  
212 the wilting-point value. Hydraulic derivation is detailed in Supplementary Section S2.

213 The 2020 ChinaCP wheat–maize class and its CPM cropland fractions defined rotation-area weights (Qiu et al.,  
214 2022). SRTM elevation supported the terrain screen and slope calculation (NASA JPL, 2013). Regional aggregation  
215 retained the common  $0.1^\circ$  meteorological grid and the study-area definition in Section 2.2.

### 216 **2.3.4. Terrestrial-water-storage observations**

217 CSR RL06.3 monthly GRACE/GRACE-FO total-water-storage anomalies were aggregated using physical cell  
218 areas and the official CSR land mask (Save, 2020; Save et al., 2016). The land-mask overlap covers 395,969 km<sup>2</sup>  
219 within the 404,658-km<sup>2</sup> study area; excluded coastal supports account for 2.15% of its area. The archive contains  
220 259 observed monthly solutions from April 2002 to July 2026 relative to the 2004–2009 mean. Source values in  
221 centimeters were converted to millimeters, and missing months remain missing. A 12-calendar-month moving mean  
222 requires at least nine observations. The 0.25° product grid is finer than the effective GRACE observation scale, so  
223 one regional storage indicator is shared across all simulation units (Save, 2020; Save et al., 2016). Soil, surface and  
224 other stores were not subtracted; the indicator represents total water storage rather than groundwater alone.

225 The corresponding aggregation without the CSR land mask is retained as a sensitivity analysis. Regional storage  
226 changes provide hydrological context for water-management pressures, with no causal attribution to particular  
227 irrigation treatments, water transfers or groundwater regulations (Long et al., 2025; Long et al., 2020).

## 228 **2.4. Crop model description**

229 Open Crop Model (OCM) couples crop growth and a layered soil-water balance at a daily time step. Wheat  
230 development combines temperature, vernalization and photoperiod responses; maize development follows thermal  
231 time. Development stage (DVS) is anchored at emergence, flowering and maturity. Germination begins when water  
232 above the wilting-point storage is available in the represented layer containing the seed; subsequent thermal  
233 development starts from that date. Stage coefficients remain fixed, while the soil-water state can alter establishment  
234 timing. Recorded or prescribed harvest dates define the management endpoint separately. Soil-water storage passes  
235 continuously through crops and fallows.

236 Canopy interception of radiation follows Beer–Lambert attenuation. Aboveground production depends on  
237 intercepted PAR, crop-specific RUE, temperature, prescribed nutrition and an empirical power response to realized  
238 transpiration supply. PAR is 0.45 of incident shortwave radiation (Kiniry et al., 1989). Stage-specific SLA  
239 determines new leaf area, while established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.).  
240 Stage-dependent thermal senescence and a separate daily stress-loss fraction determine leaf survival (Tan et al.,  
241 2020). LAI emerges from these carbon and area balances without a maximum-LAI parameter. Simulated biomass at  
242 wheat flowering or maize silking determines grain number; fixed individual grain mass defines sink capacity, and  
243 assimilate supply and wheat stem reserves constrain grain filling.

244 The water balance represents interception, ponding, infiltration, evaporation, root uptake, runoff and bottom  
245 drainage. Atmospheric demands are partitioned from reference ET using canopy cover and effective transpiration  
246 and evaporation coefficients. Soil evaporation follows a storage-based two-stage reduction (Allen et al., 1998).  
247 Root uptake depends on rooted-layer water, local depletion stress and a bounded compensation fraction that  
248 redirects unmet demand toward better-supplied rooted layers. Vertical transport uses van Genuchten–Mualem  
249 hydraulic functions (Mualem, 1976; van Genuchten, 1980), signed gravity–matric gradients and conservative  
250 storage bounds. Relative-arithmetic interface mobility and adaptive internal steps permit wetting and redistribution  
251 within 24 daily outer steps. The lower boundary is free drainage with no capillary-rise supply. This explicit layered  
252 approximation differs from a full implicit Richards-equation solution. Profile depths follow the supplied inputs,  
253 including 2 m for Wuqiao and regional simulations.

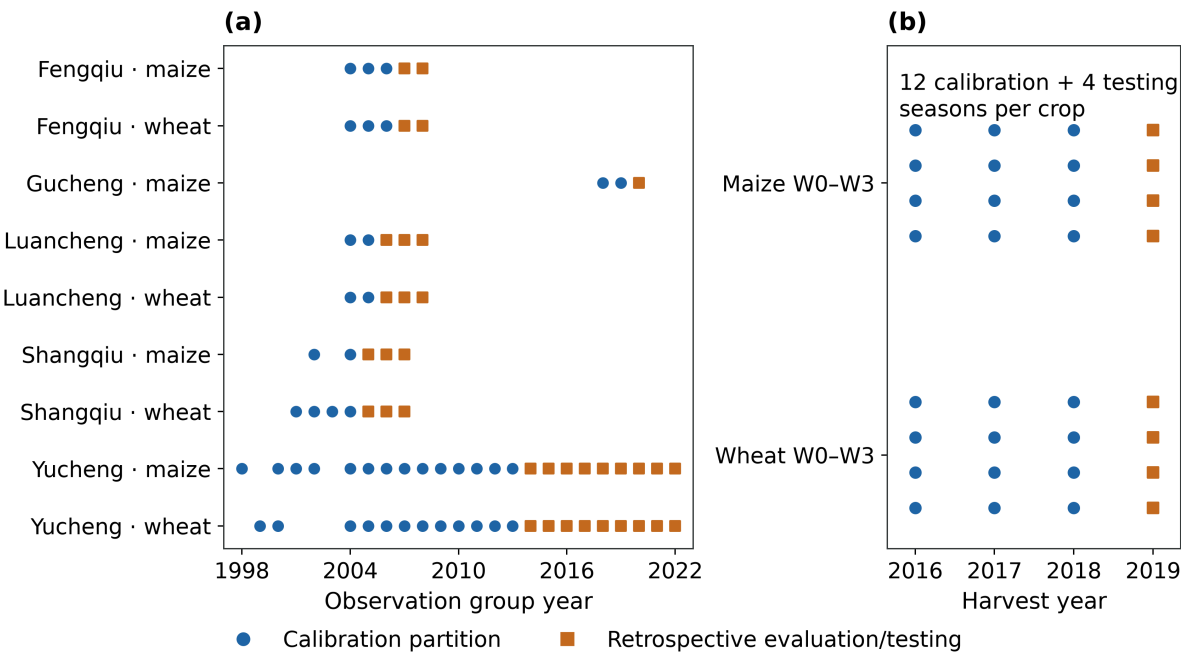
## 254 **2.5. Model parameterization, calibration and evaluation**

255 Phenological coefficients, stage-specific specific leaf area and senescence, initial LAI, stem-reserve coefficients and  
256 nutrition modifiers follow the crop parameterization described in Section S2. RUE, leaf allocation and grain-  
257 number parameters were estimated from 480 wheat and 433 maize growth and harvest observations across four  
258 wheat sites and five maize sites. Whole site-years retain their original partitions, with balanced contributions from  
259 variables, sites, site-years and crop cases. Station simulations retain their recorded site and cultivar  
260 parameterization; Wuqiao and regional simulations use the shared crop parameters (Table S3). Individual grain  
261 masses are fixed at 0.045 g for wheat and 0.300 g for maize (APSIM Initiative, 2026).

262 Eight water-use parameters per crop are fitted to the documented Wuqiao treatments in 2016–2018: the  
263 transpiration coefficient, soil evaporation coefficient, readily available water fraction, assimilation water-stress  
264 exponent, root-density decay, maximum rooting depth, root water-uptake coefficient and root compensation  
265 fraction. Each crop contributes 12 seasonal ET, 12 dry-grain and 12 dry-biomass targets, with three annual W3–W0  
266 ET contrasts derived from the same treatment seasons. RUE, leaf allocation and grain-number parameters remain  
267 fixed. Bounded least squares combines scaled observation residuals and weak parameter priors; objective weights,  
268 bounds and convergence settings are specified in Section S2.4.

269 The fitted water parameters are fixed before comparison with the four 2019 treatment means per crop. Previous  
270 inspection of these outcomes makes the test retrospective. Station observations have zero weight in this water fit  
271 and assess transfer under unconfirmed complete irrigation histories. Regional strategy selection uses historical

weather responses; the crop-calibration years overlap the 2014–2025 regional comparison period, so those outcomes are conditional scenarios rather than time-independent crop validation. Observed and simulated quantities retain matched sampling windows and grain-moisture bases.



**Figure 3. Evidence partitions. (a) Whole station site-years retain their original calibration and retrospective-evaluation partitions; dates denote observation-group years. All five eligible sites are included. (b) Four Wuqiao treatment means per crop-year provide 12 calibration seasons in 2016–2018 and four retrospective testing seasons in 2019. Station comparisons retain earlier growth-fitting history and unconfirmed complete irrigation management.**

## 2.6. Regional upscaling and irrigation scenarios

Thirty-two area-weighted climate–soil representatives were selected from actual source cells using deterministic clustering of latitude, longitude, plant-available water, and 1997–2013 mean precipitation and reference ET. Features were standardized with rotation-area weights. Each cluster uses the actual cell nearest its centroid; the full source-cell mapping and represented areas are retained. Climate years used for policy evaluation do not enter the climate features used for selection. Shared crop parameter sets are transferred regionally without extrapolating site-specific adjustments.

Each representative has five continuous histories at irrigation fractions 0, 0.25, 0.50, 0.75 and 1.00. The conventional wheat schedule is 60 mm at sowing on 12 October, 90 mm on 25 March, 90 mm on 15 April and 60 mm on 10 May, totaling 300 mm. Maize receives 80 mm at sowing on 15 June. These common schedules are declared scenarios and are not observed universal practice across the plain. Every event amount is multiplied by the irrigation fraction, with its timing fixed. Wheat is simulated through 10 June and maize through 5 October;

291 intervening fallows carry the soil state forward. Initial plant-available water is 80% at the start of the 1996 spin-up,  
292 with no subsequent seasonal storage reset. All crop and fallow ET, drainage and runoff are included. The final 2025  
293 cycle ends on 5 October because the next fallow lies outside the declared endpoint. Nutrition is represented by  
294 prescribed fertilized-crop modifiers; nitrogen cycling and spatially varying nutrient limitations are not dynamically  
295 simulated.

296 Eight purposively selected non-representative cells provide paired spatial aggregation checks under all five  
297 irrigation fractions. Two cells from each latitude quartile have the largest area-times-feature-distance score. Actual-  
298 cell weather and profiles use the same continuous model and crop parameters. Their differences from  
299 representative-cell predictions measure crop-model aggregation error rather than field-observation accuracy  
300 (Supplementary Section S3).

## 301 **2.7. Irrigation strategies and comparisons**

### 302 **2.7.1. Spatial allocation under common irrigation budgets**

303 Uniform allocation assigns the same irrigation fraction to every represented hectare. Targeted allocation  
304 chooses area shares  $x_{jk}$  for representative  $j$  and simulated irrigation fraction  $k$  by linear programming. The  
305 objective maximizes  $\sum_j A_j \sum_k x_{jk} \bar{Y}_{jk}$ , where  $\bar{Y}_{jk}$  is mean annual wheat-plus-maize grain in 1997–2013.  
306 Constraints require  $\sum_k x_{jk} = 1$ ,  $0 \leq x_{jk} \leq 1$  and  $\sum_j A_j \sum_k x_{jk} I_k = (1 - r) \sum_j A_j I_1$ , with reductions  
307  $r = 0.25, 0.50$  or  $0.75$ . Within-representative mixtures describe shares of area receiving actual simulated  
308 irrigation treatments; crop responses are not interpolated between irrigation fractions. Allocations remain  
309 fixed in 2014–2025. Both policies deliver exactly the same total field-irrigation water in every year, so  
310 their differences reflect allocation under the same volume constraint.

311 Spatial contrasts at the common 50% irrigation cut use frozen shares of complete continuous irrigation histories.  
312 Their area-weighted grain, ET and drainage differences map from 32 representatives to 3,641 source cells. Fixed  
313 southern (32–36°N), central (36–38°N) and northern (38–41°N) bands summarize the distribution. Counts of  
314 positive annual contrasts describe consistency within the modeled record (Figure 7; Table S9).

### 315 **2.7.2. Annual quotas based on antecedent water availability**

316 Relative water-availability classes combine two antecedent indicators: regional total-water-storage status and local  
317 rainfall before wheat sowing. The storage indicator is the mean of available August and September GRACE  
318 observations in the calendar year preceding harvest, with at least one monthly solution required. Precipitation is  
319 accumulated from 1 July to 30 September at each simulation unit, covering 92 days before the prescribed 12  
320 October wheat sowing. The corresponding precipitation/reference-ET ratio is retained as a diagnostic of climatic  
321 water availability. These antecedent indicators do not use rainfall or crop outcomes from the subsequent growing  
322 season.

323 Thresholds use 2003–2013 harvest years, the common selection period with antecedent GRACE coverage. The  
324 regional storage median is 13.38 mm, and the rotation-area-weighted precipitation median is 431.74 mm. Values  
325 below their respective thresholds define lower storage and lower rainfall; values at or above define higher storage  
326 and higher rainfall. Crossing the two indicators gives four hydroclimatic classes. The classes describe relative  
327 antecedent water availability, rather than absolute water scarcity derived from water demand, allocation rights or  
328 access. Spatial differences within a year arise from precipitation; GRACE does not supply independently resolved  
329 field-scale storage classes.

330 For each class, five continuous irrigation histories provide paired grain and ET responses relative to conventional  
331 irrigation, weighted by rotation area and year. The primary candidate is the lowest fraction retaining at least 95% of  
332 class-mean conventional grain in 2003–2013; a 98% target assesses a stricter production criterion. Selected  
333 fractions and class thresholds remain frozen in 2014–2025. Higher-storage classes have no evaluation exposure.  
334 Required GRACE observations are missing for 2014, 2018 and 2019.

335 Two annual strategies apply the irrigation fractions selected from fixed-fraction responses at grain-retention targets  
336 of 95% and 98%. At each simulation unit, the antecedent storage–rainfall class determines the fraction before wheat  
337 sowing. This fraction scales all four wheat applications and the subsequent maize application in the same rotation  
338 year. Application dates, crop parameters, weather, hydraulic profiles and the free-drainage lower boundary are  
339 identical to those in the fixed-fraction experiment. Conventional irrigation applies before 2003 and whenever an  
340 eligible storage class is unavailable, including evaluation harvest years 2014, 2018 and 2019.

341 Each strategy is simulated continuously from the 1996 spin-up through 2025. Soil-water states pass between crops  
342 and fallows and across annual changes in irrigation fraction. Conventional irrigation supplies the paired continuous  
343 reference; replays at two simulation units verify the scenario driver against conventional outputs. Class thresholds  
344 and irrigation fractions remain fixed after selection. Outcomes cover all twelve comparison years and, separately,

the nine years with eligible classes. Production retention is the ratio of period-mean regional grain production to its corresponding conventional mean; annual ratios describe variation within the period. The strategies assume access to antecedent observation periods, while release availability of GRACE before sowing remains unverified.

### **2.7.3. Rainfall-only controls and matched monitoring coverage**

Rainfall-only controls use the same 2003–2013 simulation-unit years with eligible GRACE observations, the same rotation-area weights, irrigation candidates and grain-retention targets. Selection groups responses by antecedent rainfall alone and chooses the lowest fraction meeting the relevant target. The rainfall threshold is identical to that in the joint classification. Matched controls receive conventional irrigation whenever the storage–rainfall strategy lacks an eligible class, preserving both the selection cohort and monitoring coverage. Continuous differences between these paired strategies consequently assess the incremental effect of storage-based grouping.

Separate rainfall-available controls apply the selected rainfall-only fractions whenever an eligible rainfall group exists, including years without GRACE coverage. Their complete histories assess the combined effect of rainfall-based management and greater monitoring availability. Irrigation changes in earlier gap years can alter the soil-water state of later crops, so these controls are reported separately from the matched comparison.

## **2.8. Evaluation metrics, spatial statistics and water accounting**

Performance is summarized by RMSE, mean bias error, nRMSE relative to observed mean, Nash–Sutcliffe efficiency (NSE) (Nash and Sutcliffe, 1970) and squared Pearson correlation ( $R^2$ ). Station scores give equal weight to available sites, site-years within sites, crop cases within site-years and records within cases. Wuqiao treatment means receive equal weights within each crop and partition. Field calibration and retrospective testing remain separate. Grain and harvest biomass in station comparisons are dry matter; Wuqiao grain results retain the published 13% moisture basis, with a single 0.87 conversion between dry and reported grain. Biomass remains dry matter throughout.

Seasonal curves group matching observations and predictions into 15-day wheat and 7-day maize bins at observation-window midpoints in days after sowing. Case means receive equal weight within site-years, site-year means within sites and available sites within bins. Standard errors describe variation among contributing site-year means rather than model uncertainty. Site-specific curves use equal site-year weights. Coverage varies among bins, and gaps remain uninterpolated. Accuracy scores use unbinned records. Yucheng harvest error bars describe quadrat sampling variation within crop-years.

ET sums use identical observed and simulated measurement days within each case and retain the station balancing weights. They exclude unmeasured days and do not represent complete-season observed totals. Regional grain production uses dry mass and rotation-area weights; actual ET includes crops and fallows. Paired policy differences cover 2014–2025. Daily water and carbon residuals and continuity of carried soil-water storage assess numerical conservation. Field irrigation, ET, drainage below 2 m and groundwater storage have distinct accounting boundaries.

Spatial distribution statistics describe targeted-minus-uniform policy contrasts in mean annual combined wheat–maize dry grain and water fluxes over 2014–2025. The 3,641 mapped 0.1° source cells carry outcomes from 32 representative response units. Cell weights are the static 2020 mapped rotation areas after cropland-fraction correction; they represent mapped rotation area rather than observed harvested area. Gains and losses exceed  $+10^{-6}$  and fall below  $-10^{-6}$  kg ha<sup>-1</sup>, respectively; contrasts within these bounds are classified as unchanged. Group means use rotation-area weights, regional contributions sum area multiplied by the corresponding contrast, and weighted quantiles use the inverse empirical cumulative area distribution. Annual loss frequencies use the same sign threshold across the 12 years. Connected loss footprints comprise occupied cells sharing a horizontal or vertical grid edge (rook adjacency). The statistics describe the conditional modeled allocation footprint; mapped cells do not provide independent replication or cell-specific validation.

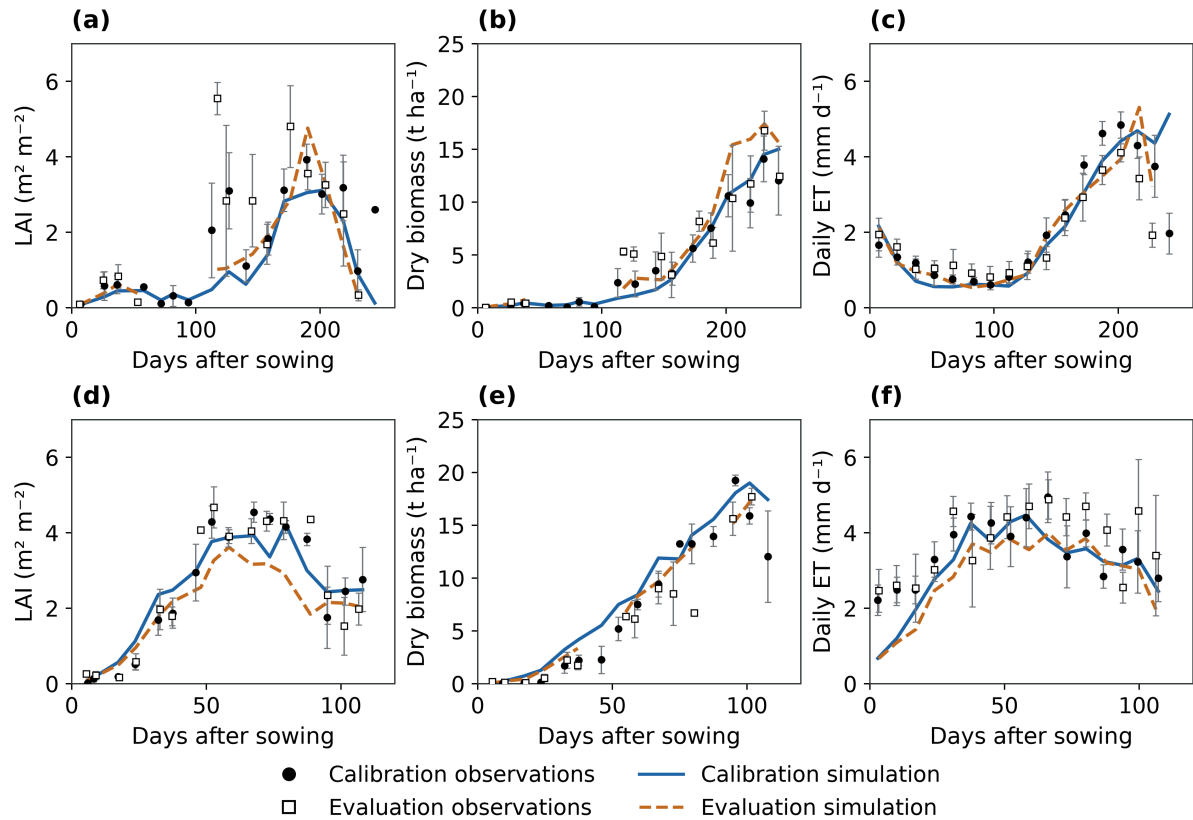
### 3. Results

#### 3.1. Conditional multisite crop comparisons

Retrospective-evaluation LAI RMSE was 1.843 m<sup>2</sup> m<sup>-2</sup> for wheat and 0.995 m<sup>2</sup> m<sup>-2</sup> for maize, with NSE of 0.184 and 0.689, respectively (Figure 4). LAI RMSE in the original calibration partition was 1.727 and 1.138 m<sup>2</sup> m<sup>-2</sup>. In-season biomass RMSE in retrospective evaluation was 4.413 and 2.789 t ha<sup>-1</sup>, with biases of +1.314 and +1.158 t ha<sup>-1</sup> and NSE of 0.349 and 0.796. Site-specific seasonal comparisons retain their available observations and original partitions (Figures S6–S11).

Dry-grain RMSE was 1.256 t ha<sup>-1</sup> for wheat and 1.402 t ha<sup>-1</sup> for maize in the original calibration partition, compared with 0.834 and 2.286 t ha<sup>-1</sup> in retrospective evaluation (Figure S5). Evaluation grain biases were +0.485 and -1.586 t ha<sup>-1</sup>, and NSE was 0.099 and -1.566. Evaluation harvest-biomass RMSE was 2.736 t ha<sup>-1</sup> for wheat and 3.719 t ha<sup>-1</sup> for maize; the corresponding observation counts were 54 in nine wheat years and 48 in eight maize years.

401 Daily ET RMSE in retrospective evaluation was 1.809 mm d<sup>-1</sup> for wheat and 2.572 mm d<sup>-1</sup> for maize, with biases  
 402 of +0.141 and -0.734 mm d<sup>-1</sup> and NSE of 0.121 and 0.012 (Figure 4). ET summed over matched measurement days  
 403 had evaluation RMSE of 51.29 mm for wheat and 74.17 mm for maize, biases of +25.94 and -55.58 mm, and NSE  
 404 of 0.826 and 0.118 (Table 2). These sums exclude days without ET measurements.



405 **Figure 4. Multisite seasonal crop comparisons. Wheat and maize occupy the top and bottom rows; columns show LAI, aboveground**  
 406 **dry biomass and daily actual ET. Filled and open observations distinguish the original calibration and retrospective-evaluation**  
 407 **partitions; solid blue and dashed orange lines show the corresponding simulations. Error bars describe standard errors among site-**  
 408 **year means. Balanced means use 15-day wheat and seven-day maize bins, and unobserved bins remain gaps. Accuracy scores use**  
 409 **unbinned records. Complete station irrigation management remains unconfirmed.**

410 **Table 2. Conditional station ET sums on matched measurement days.**

| Crop  | Dataset                        | Seasons | NSE    | RMSE (mm) | MBE (mm) |
|-------|--------------------------------|---------|--------|-----------|----------|
| Maize | Prior calibration<br>partition | 11      | 0.511  | 45.60     | -34.92   |
| Maize | Retrospective<br>evaluation    | 10      | 0.118  | 74.17     | -55.58   |
| Wheat | Prior calibration<br>partition | 11      | -0.300 | 84.94     | -23.53   |
| Wheat | Retrospective<br>evaluation    | 11      | 0.826  | 51.29     | 25.94    |

411 Observed and simulated sums use identical measurement days within cases, with balanced site, site-year and case  
 412 weights. Unmeasured days are excluded. Original calibration partitions retain an earlier growth-fitting history;  
 413 evaluation is retrospective. Complete irrigation-log matching remains unconfirmed.

### 414 **3.2. Wuqiao calibration and retrospective testing**

415 Seasonal ET calibration RMSE was 40.19 mm for wheat and 37.44 mm for maize across 12 treatment seasons per  
 416 crop, with biases of -12.65 and +3.60 mm and nRMSE of 8.97% and 12.69% (Table 3; Figure 5). Across the four  
 417 2019 testing means per crop, ET RMSE was 73.80 and 55.64 mm, bias was +48.93 and -35.90 mm, nRMSE was  
 418 21.98% and 14.13%, and NSE was 0.503 and -0.346, respectively.

419 At 13% grain moisture, calibration RMSE was 1.249 t ha<sup>-1</sup> for wheat and 2.267 t ha<sup>-1</sup> for maize; retrospective-  
 420 testing RMSE was 3.161 and 1.700 t ha<sup>-1</sup>. Testing grain biases were +3.108 and +1.403 t ha<sup>-1</sup>, with nRMSE of  
 421 83.27% and 15.97% and NSE of -4.654 and -3.641 (Table 3; Figure 5). Dry aboveground-biomass RMSE was  
 422 4.222 and 4.688 t ha<sup>-1</sup> in calibration and 5.427 and 2.038 t ha<sup>-1</sup> in testing, respectively. Testing biomass nRMSE  
 423 was 55.66% for wheat and 10.44% for maize.

424 W3-W0 seasonal ET-contrast RMSE was 45.79 mm for wheat and 66.84 mm for maize across three calibration  
 425 years; the absolute 2019 contrast errors were 116.37 and 52.74 mm. For grain contrasts of W1-W3 relative to same-  
 426 year W0, calibration RMSE was 0.519 and 1.669 t ha<sup>-1</sup> and retrospective-testing RMSE was 1.316 and 1.323 t ha<sup>-1</sup>,  
 427 respectively, at 13% moisture (Table S5; Figure S2). Maize treatment labels denote preceding wheat irrigation;  
 428 maize irrigation was identical within each year.

429 **Table 3. Wuqiao water calibration and retrospective testing.**

| Crop  | Period    | Target      | n  | Unit               | RMSE   | MBE     | nRMSE<br>(%) | NSE    | R <sup>2</sup> |
|-------|-----------|-------------|----|--------------------|--------|---------|--------------|--------|----------------|
| Maize | 2016–2018 | Seasonal ET | 12 | mm                 | 37.444 | 3.604   | 12.69        | 0.533  | 0.619          |
| Maize | 2016–2018 | Grain       | 12 | t ha <sup>-1</sup> | 2.267  | -1.738  | 18.78        | -2.940 | 0.128          |
| Maize | 2016–2018 | Biomass     | 12 | t ha <sup>-1</sup> | 4.688  | -4.188  | 21.79        | -5.625 | 0.073          |
| Maize | 2019      | Seasonal ET | 4  | mm                 | 55.637 | -35.902 | 14.13        | -0.346 | 0.343          |
| Maize | 2019      | Grain       | 4  | t ha <sup>-1</sup> | 1.700  | 1.403   | 15.97        | -3.641 | 0.972          |
| Maize | 2019      | Biomass     | 4  | t ha <sup>-1</sup> | 2.038  | 0.726   | 10.44        | -0.313 | 0.123          |
| Wheat | 2016–2018 | Seasonal ET | 12 | mm                 | 40.188 | -12.649 | 8.97         | 0.689  | 0.758          |
| Wheat | 2016–2018 | Grain       | 12 | t ha <sup>-1</sup> | 1.249  | -0.445  | 16.74        | 0.746  | 0.817          |
| Wheat | 2016–2018 | Biomass     | 12 | t ha <sup>-1</sup> | 4.222  | -2.480  | 25.48        | 0.496  | 0.781          |
| Wheat | 2019      | Seasonal ET | 4  | mm                 | 73.805 | 48.929  | 21.98        | 0.503  | 0.915          |
| Wheat | 2019      | Grain       | 4  | t ha <sup>-1</sup> | 3.161  | 3.108   | 83.27        | -4.654 | 0.870          |
| Wheat | 2019      | Biomass     | 4  | t ha <sup>-1</sup> | 5.427  | 4.908   | 55.66        | -0.752 | 0.924          |

430 Calibration uses 12 treatment seasons per crop in 2016–2018; retrospective testing uses four in 2019. Counts are  
431 treatment means rather than independent replicate seasons. ET is a 0–2 m water-balance estimate under negligible  
432 runoff and drainage assumptions. Grain is at 13% moisture and aboveground biomass is dry matter. Annual grain  
433 means and SE are digitized from source Supplementary Figure S2. Initial total storage is observed; layer  
434 distribution and transferred hydraulics remain conditional.

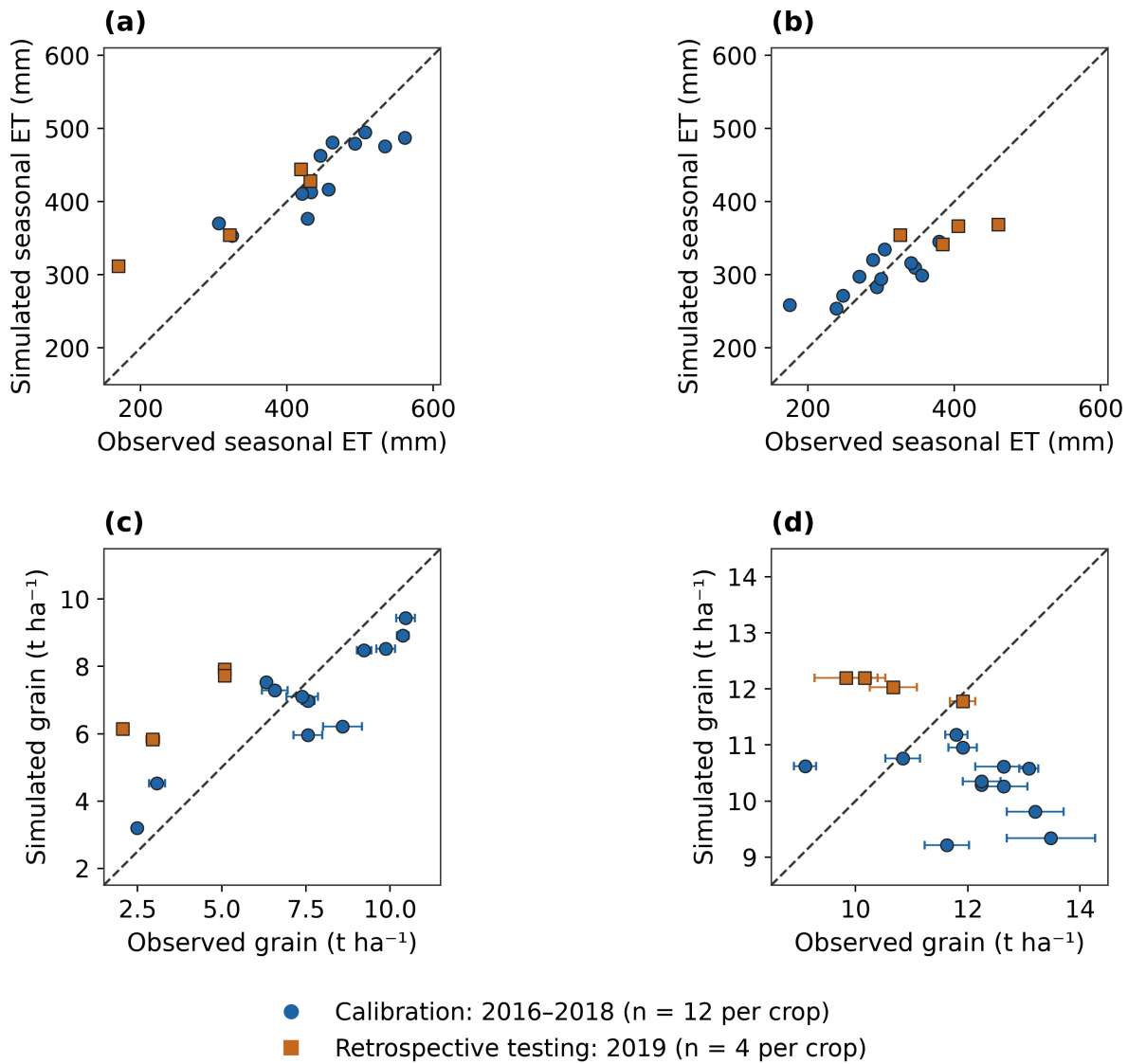


Figure 5. Wuqiao calibration and retrospective testing. (a,b) Seasonal ET; (c,d) annual treatment-mean grain at 13% moisture. Wheat occupies the left column and maize the right. Circles show 2016–2018 treatments used to fit water-use parameters; squares show 2019, withheld from the fit and selection. Dashed lines denote 1:1 agreement. Crop-specific grain axes focus on their observed ranges. Horizontal grain error bars are the published SE from three field replicates. Observed ET follows the published 0–2 m water balance; simulated dry grain is converted once by division by 0.87. Maize labels retain preceding wheat management with identical maize irrigation within each year.

### 3.3. Grain production and water balance under irrigation allocation

At a 50% regional irrigation reduction, targeted allocation gave a simulated annual rotation dry-grain yield of 15.761 t ha<sup>-1</sup> yr<sup>-1</sup> for wheat and maize combined, compared with 15.029 under uniform cuts. Both yields remained below the conventional yield of 16.756 t ha<sup>-1</sup> yr<sup>-1</sup>. Targeting therefore retained 0.732 t ha<sup>-1</sup> yr<sup>-1</sup> more grain than

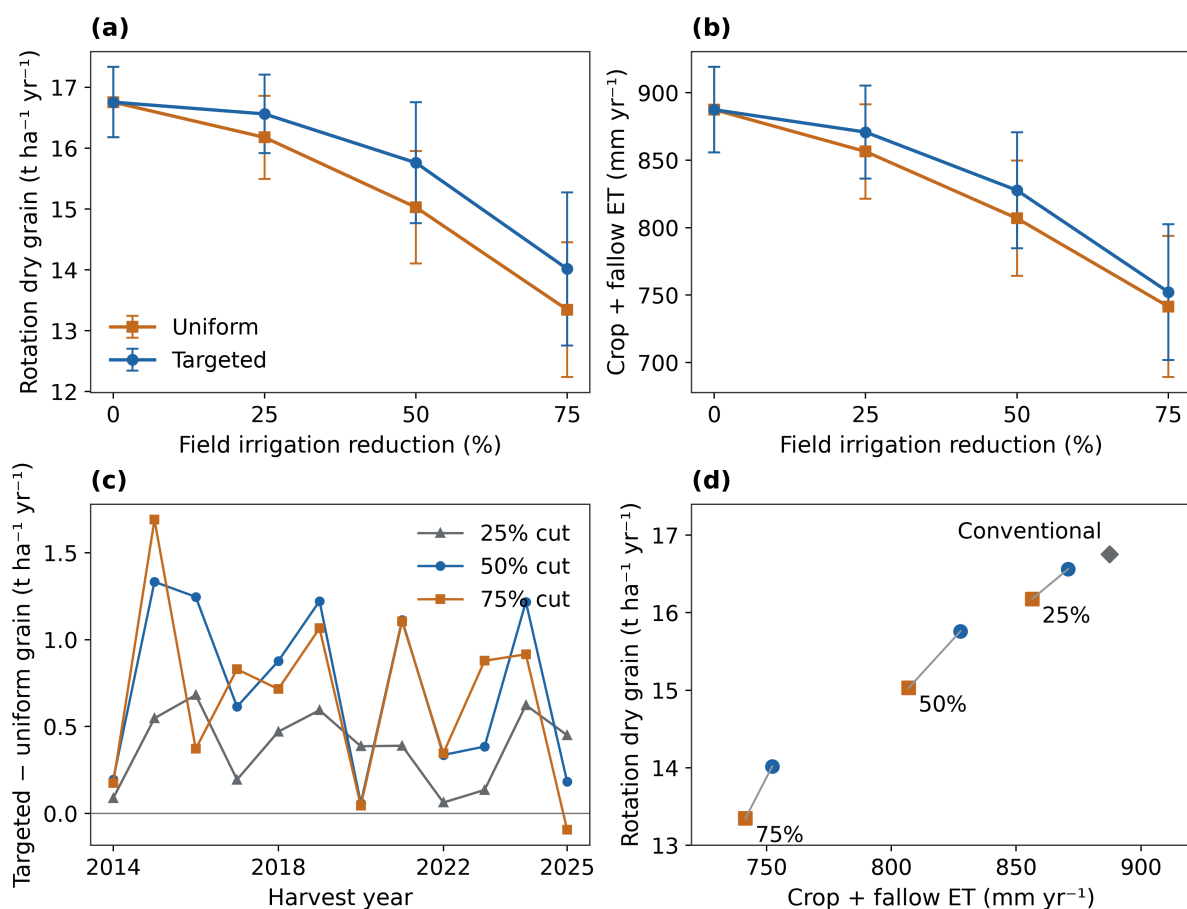
uniform allocation (4.87% of uniform production), equivalent to 7.219 million t yr<sup>-1</sup> across the fixed 9.867 million rotation hectares (Table 4; Figure 6). Targeted production remained 9.813 million t yr<sup>-1</sup> below conventional production.

Both allocations supplied 190 mm yr<sup>-1</sup> of field irrigation. Targeted and uniform crop-plus-fallow ET averaged 827.70 and 806.94 mm yr<sup>-1</sup>, respectively. The grain advantage was accompanied by 20.76 mm yr<sup>-1</sup> (2.048 km<sup>3</sup> yr<sup>-1</sup>) greater ET. Profile drainage was 79.04 versus 96.34 mm yr<sup>-1</sup>, and runoff was 14.43 versus 18.99 mm yr<sup>-1</sup> (Table 4). The targeted-minus-uniform grain contrast at the 50% cut was positive in all twelve comparison years (Figure 6c). The positive regional contrast was accompanied by spatially contrasting yield responses (Section 3.4).

**Table 4. Simulated annual rotation production and water balance under equal irrigation reductions, 2014–2025.**

| Strategy          | Dry grain (t ha <sup>-1</sup> yr <sup>-1</sup> ) | Irrigation (mm yr <sup>-1</sup> ) | ET (mm yr <sup>-1</sup> ) | Drainage (mm yr <sup>-1</sup> ) | Runoff (mm yr <sup>-1</sup> ) |
|-------------------|--------------------------------------------------|-----------------------------------|---------------------------|---------------------------------|-------------------------------|
| Conventional      | 16.756                                           | 380.00                            | 887.43                    | 185.26                          | 39.32                         |
| Uniform, 25% cut  | 16.177                                           | 285.00                            | 856.41                    | 135.94                          | 25.71                         |
| Targeted, 25% cut | 16.562                                           | 285.00                            | 870.81                    | 125.90                          | 19.86                         |
| Uniform, 50% cut  | 15.029                                           | 190.00                            | 806.94                    | 96.34                           | 18.99                         |
| Targeted, 50% cut | 15.761                                           | 190.00                            | 827.70                    | 79.04                           | 14.43                         |
| Uniform, 75% cut  | 13.347                                           | 95.00                             | 741.57                    | 68.72                           | 15.83                         |
| Targeted, 75% cut | 14.018                                           | 95.00                             | 752.30                    | 60.01                           | 13.68                         |

All per-hectare means use the same 9.866602 million mapped rotation hectares. Dry grain combines wheat and maize harvests; ET includes crops and fallows. Regional grain in million tonnes equals rotation yield multiplied by mapped area in million hectares. Regional water in km<sup>3</sup> equals depth in mm multiplied by hectares and 10, divided by 10<sup>9</sup>. The conventional schedule supplies 300 mm to wheat and 80 mm to maize.



458 **Figure 6. Irrigation allocation and grain–water trade-offs during 2014–2025. (a,b) Rotation-area-weighted annual combined wheat–**  
 459 **maize dry grain and crop-plus-fallow ET; error bars are between-year SD. (c) Annual targeted-minus-uniform grain at identical**  
 460 **irrigation budgets. (d) Period means; connectors join policies with the same budget. Values per hectare use the same 9.867 million**  
 461 **mapped rotation hectares, rather than separate crop areas. Targeting retains more grain than uniform cuts while remaining below**  
 462 **conventional production.**

### 463 3.4. Spatial distribution and persistence of yield contrasts

464 Relative to uniform allocation, the central and northern latitude bands contributed grain gains of 7.475 and 1.886  
 465 million  $\text{t yr}^{-1}$ , while the southern band contributed a loss of 2.142 million  $\text{t yr}^{-1}$ . The southern band occupied 60.7%  
 466 of mapped rotation area. Mean rotation-yield contrasts were +2.415, +2.420 and  $-0.357 \text{ t ha}^{-1} \text{yr}^{-1}$  in the central,  
 467 northern and southern bands, respectively (Figure 7; Table S9).

468 At the 50% regional irrigation reduction, targeted allocation increased mean annual combined wheat–maize dry  
 469 grain relative to uniform reduction on 43.92% of the mapped rotation area (4.333 Mha), decreased it on 37.60%  
 470 (3.710 Mha), and left it unchanged on 18.48% (1.823 Mha). Mean contrasts within the gain and loss areas were

471 +2.439 and -0.903 t ha<sup>-1</sup> yr<sup>-1</sup>, respectively. Their regional contributions were +10.568 and -3.349 Mt yr<sup>-1</sup>, giving a  
472 net regional grain difference of 7.219 Mt yr<sup>-1</sup> (0.732 t ha<sup>-1</sup> yr<sup>-1</sup> across the full mapped area). Local losses offset  
473 31.69% of the gross positive contribution (Figure 7d).

474 The area-weighted median grain contrast was 0.000 t ha<sup>-1</sup> yr<sup>-1</sup>, and the 10th and 90th percentiles were -1.182 and  
475 +3.553 t ha<sup>-1</sup> yr<sup>-1</sup>. Relative to uniform reduction, mean annual field irrigation increased by 119.0 mm yr<sup>-1</sup> in the  
476 gain area and decreased by 139.0 mm yr<sup>-1</sup> in the loss area. The corresponding crop-plus-fallow ET contrasts were  
477 +94.4 and -55.1 mm yr<sup>-1</sup>. Irrigation changes balanced at regional scale, whereas the area-weighted mean ET  
478 contrast was +20.8 mm yr<sup>-1</sup>.

479 Negative yield contrasts persisted over 2014–2025: 21.34% of the mapped rotation area had lower yields than under  
480 uniform allocation in all 12 years, and 37.60% experienced a loss in at least one year. Every response unit with a  
481 negative period-mean contrast recorded losses in at least 9 years; all response units with a positive period-mean  
482 contrast recorded gains in all 12 years. Across years, the mean annual area exposed to a loss was 35.37%. Mapped  
483 cells with negative period-mean contrasts formed 8 connected components under rook adjacency. The largest  
484 contained 3.700 Mha of mapped rotation area (99.73% of all loss area), with occupied cell centers extending from  
485 32.0 to 36.3°N and 112.8 to 121.5°E.

486 Negative period-mean yield contrasts occupied 32.80%, 37.60%, and 60.08% of mapped rotation area at reductions  
487 of 25%, 50%, and 75%, respectively. At the 75% reduction, the area-weighted median contrast became negative  
488 (-0.788 t ha<sup>-1</sup> yr<sup>-1</sup>), although gains on 37.04% of the area produced a positive regional balance. Gross positive and  
489 negative contributions at that reduction were +14.100 and -7.487 Mt yr<sup>-1</sup>, leaving a net gain of 6.614 Mt yr<sup>-1</sup>, below  
490 the net gain at the 50% reduction.

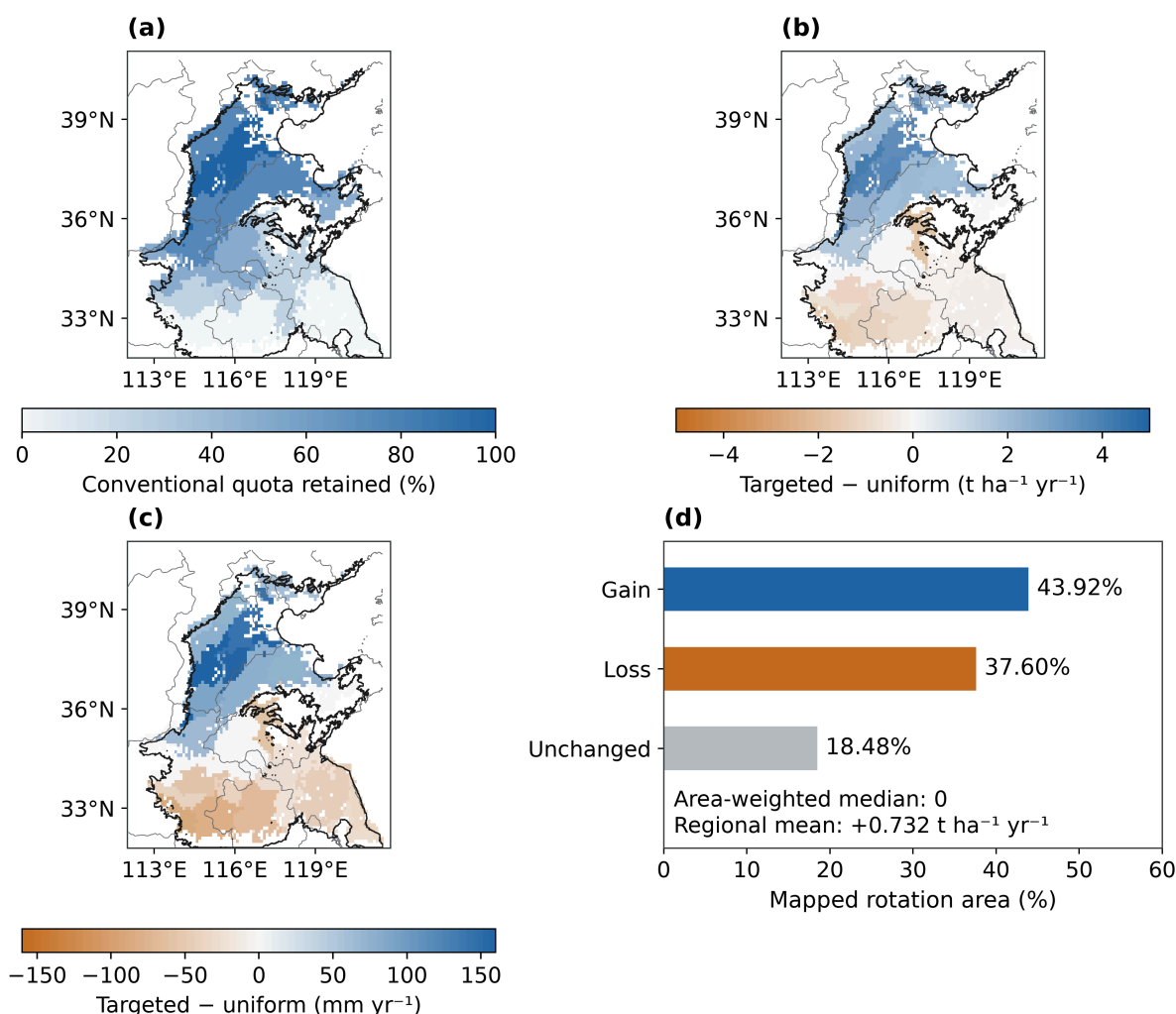


Figure 7. Spatial redistribution at a common 50% regional irrigation reduction, 2014–2025. (a) Targeted fraction of the conventional quota; uniform allocation retains 50% everywhere. (b,c) Targeted-minus-uniform combined rotation dry grain and crop-plus-fallow ET. (d) Mapped rotation-area shares with positive, negative or unchanged mean grain contrasts. The 3,641 cells inherit 32 representative responses. Thin grey provincial lines and black study-domain outlines follow Figure 2; unmapped areas remain blank.

### 3.5. Antecedent water availability and annual irrigation strategies

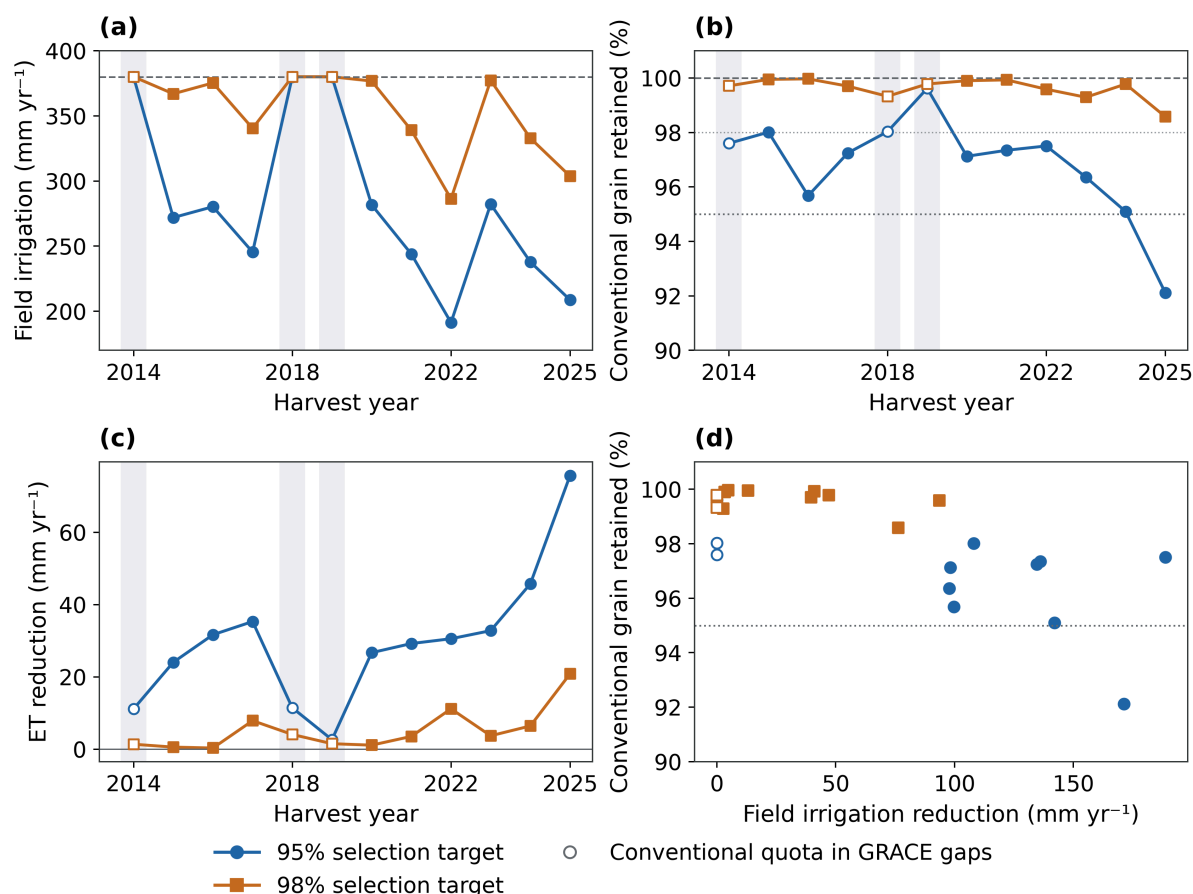
Four antecedent storage–rainfall classes occurred during 2003–2013. Nine comparison years had eligible GRACE observations, all in the lower-storage category; three years without eligible GRACE observations received conventional irrigation quotas. At the 95% grain target, both storage strata selected an irrigation fraction of 0.75 under lower rainfall and 0.50 under higher rainfall. The 98% target selected 1.00 and 0.75, respectively (Table 5). Continuous strategies selected at the 95% and 98% targets retained 96.27% and 99.63% of conventional grain production over the nine years with eligible GRACE observations. Field-irrigation reductions were 130.72 and 35.72 mm per rotation; corresponding ET reductions were 36.84 and 6.17 mm (Figure 8; Table S7). The 95% strategy retained at least 95% in 8 of nine years, with a minimum of 92.12%.

504 Storage–rainfall and matched rainfall-only strategies selected identical irrigation fractions and produced identical  
 505 grain, ET, drainage and irrigation histories at both targets. Rainfall-only strategies that applied decisions during  
 506 GRACE gaps followed different irrigation histories and carried different soil-water states into subsequent crops  
 507 (Table S8).

508 **Table 5. Selected irrigation candidates by relative water-availability class.**

| Class              | Irrigation fraction | Wheat irrigation<br>(mm) | Maize irrigation<br>(mm) | Selection grain<br>retention (%) |
|--------------------|---------------------|--------------------------|--------------------------|----------------------------------|
| Higher storage dry | 0.750               | 225.000                  | 60.000                   | 95.537                           |
| Higher storage wet | 0.500               | 150.000                  | 40.000                   | 97.093                           |
| Lower storage dry  | 0.750               | 225.000                  | 60.000                   | 96.617                           |
| Lower storage wet  | 0.500               | 150.000                  | 40.000                   | 97.431                           |

509 Fractions meet the 95% class-mean grain target during selection years 2003–2013. Current shared crop parameters  
 510 apply throughout. Both storage strata select identical fractions within each rainfall group; higher-storage classes  
 511 have no later-period exposure.



**Figure 8. Continuous annual irrigation strategies during 2014–2025. (a) Field irrigation; (b) grain retained relative to conventional irrigation; (c) crop-plus-fallow ET reduction; (d) annual production–irrigation combinations. Blue circles and orange squares identify 95% and 98% selection targets. Shading and open markers identify missing-GRACE years with conventional quotas. Grain and ET can still differ in those years because preceding strategy decisions alter carried soil-water states. Matched rainfall-only histories coincide with the storage–rainfall histories.**

## 4. Discussion

We used a calibrated crop model to simulate continuous winter wheat–summer maize rotations across the North China Plain, compare uniform and spatially targeted irrigation under equal water budgets, and test whether antecedent regional water storage improved decisions based on rainfall. Sustaining grain production under restricted irrigation requires consideration of both consumptive water use and the distribution of production losses across locations. The conditional results show that higher regional grain retention can accompany greater evapotranspiration and lower production in some locations, whereas storage information did not change the selected irrigation quotas within the tested design. The management value of these strategies therefore depends on accounting for the complete rotation water balance, soil-water carry-over across crops and fallows, and reliable crop responses to irrigation.

#### 527 **4.1. Marginal irrigation returns and spatial allocation**

528 Spatial targeting exploits differences in marginal grain response to irrigation. Diminishing production gains across  
529 field schedules (Yang et al., 2022) and spatial variation in irrigation requirements (Wu et al., 2019) provide a basis  
530 for redistributing a limited supply, consistent with regional allocation studies (Jiang et al., 2016). The equal-budget  
531 comparison attributes the simulated production difference to delivery allocation. Complete continuous histories  
532 retain the effects of earlier irrigation and rainfall on subsequent crop responses.

533 At the 50% delivery reduction, targeting retained  $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$  more combined wheat–maize dry grain than  
534 uniform cuts. Across 9.8666 million rotation hectares, the corresponding reduction in grain loss was 7.219 million t  
535  $\text{yr}^{-1}$ , or 4.87% of the uniform total. Targeted production remained below conventional production. The positive  
536 contrast in all twelve years indicates consistent regional gains within the simulated record. Its magnitude remains  
537 dependent on the crop-response functions and declared management schedules (Table 4; Figure 6).

538 The spatial distribution sharpens this distinction between regional and local outcomes. Relative to uniform  
539 allocation, mean losses occupy 37.60% of mapped rotation area, and losses persist in every comparison year on  
540 21.34%. Gross gains of  $10.568 \text{ Mt yr}^{-1}$  are partly offset by losses of  $3.349 \text{ Mt yr}^{-1}$ ; the area-weighted median is  
541 zero despite a positive regional mean. The regional grain objective balances gains and losses among locations.  
542 Local production floors constitute an additional allocation constraint.

543 Regional gain has a pronounced distributional cost. Central and northern band-mean rotation yields increased by  
544  $2.415$  and  $2.420 \text{ t ha}^{-1} \text{ yr}^{-1}$ , while the southern mean decreased by  $0.357 \text{ t ha}^{-1} \text{ yr}^{-1}$ . The southern band covers  
545 60.7% of the rotation area, so its losses substantially offset gains elsewhere (Figure 7; Table S9). The production  
546 objective maximizes the area-weighted regional total and supplies no minimum yield for an individual location.  
547 Local production floors and constraints on the distribution of irrigation reductions are therefore separate  
548 requirements for allocation that protects affected farms.

#### 549 **4.2. Rotation carry-over and the timing of water supply**

550 The spatial production response integrates water effects across both crops. Wheat irrigation can support canopy  
551 growth and grain formation, replenish water available to maize or increase drainage. Reduced wheat delivery can  
552 increase subsequent dependence on summer rainfall, producing different annual grain responses under different  
553 rainfall sequences. The sixteen-year irrigation experiment shows diminishing production gains, and the Wuqiao

554 experiment links precipitation conditions with wheat scheduling and subsequent maize production (Yang et al.,  
555 2024; Yang et al., 2022). These coupled responses make the complete rotation the relevant unit for irrigation  
556 allocation.

557 Continuous soil-water histories preserve the delayed benefits and costs of allocation and annual quota changes  
558 (Figure S3). Production can be maintained by drawing down profile storage, with later rainfall or irrigation  
559 determining its replenishment. A finite-period delivery reduction can therefore include a contribution from stored  
560 soil water. Crop-plus-fallow accounting also captures evaporation and drainage between harvest and sowing. Their  
561 inclusion connects apparent production retention with the water remaining for the next crop and the complete  
562 rotation water balance.

563 The field evidence constrains only part of this carry-over mechanism. Wuqiao simulations use reported total profile  
564 storage with an assumed layer distribution. Wheat crop-start storage follows the pre-sowing simulation, whereas  
565 maize is independently initialized from treatment-specific observations. Maize irrigation is identical among  
566 treatments within each year, and its treatment labels identify preceding wheat schedules (Yang et al., 2024). Maize  
567 comparisons thus evaluate crop responses to the initial state associated with wheat management (Figure 5; Figure  
568 S2). Regional carry-over additionally depends on the model's continuous propagation and vertical redistribution of  
569 water between crops.

### 570 **4.3. Production, consumptive use and groundwater accounting**

571 The production benefit of targeting carries a consumptive-use trade-off. At the common 50% delivery reduction,  
572 targeting increased ET by  $2.048 \text{ km}^3 \text{ yr}^{-1}$  and reduced profile drainage by  $1.707 \text{ km}^3 \text{ yr}^{-1}$  relative to uniform  
573 allocation. With rainfall and total irrigation held constant, differences in ET are balanced by drainage, runoff and  
574 soil-storage change (Table 4). A shared delivery constraint therefore permits different consumptive outcomes.  
575 Irrigation productivity and consumptive-water savings require distinct objectives (Grafton et al., 2018; Perry et al.,  
576 2009).

577 The soil-profile balance also has a different boundary from the aquifer balance. Drainage below 2 m enters a deeper  
578 vadose zone, with storage and transit outside the modeled profile (Wu et al., 2023). Reduced field delivery or  
579 profile drainage therefore provides insufficient information to quantify changes in pumping, aquifer recharge or  
580 groundwater storage. Production and ET comparisons can identify agronomic trade-offs, while groundwater  
581 outcomes require the corresponding subsurface and abstraction accounting.

582 GRACE offers regional storage context at a broader scale. Its anomalies integrate groundwater with other terrestrial  
583 stores, and its effective observation scale exceeds the displayed grid resolution (Save et al., 2016; Scanlon et al.,  
584 2016). Recent aquifer recovery reflects water transfers, abstraction management and hydroclimatic conditions  
585 (Long et al., 2025; Xu et al., 2026). The crop-model scenarios quantify changes in field delivery, consumption and  
586 profile outflow without attributing those observed storage changes to individual irrigation strategies.

#### 587 **4.4. When storage information changes an irrigation decision**

588 A storage indicator adds decision value when its inclusion changes the selected action or improves the grain–water  
589 outcome under common constraints. GRACE represents regional storage conditions, while antecedent rainfall  
590 describes more local water input (Save et al., 2016; Scanlon et al., 2016). Their different physical support motivates  
591 joint classification, but decision value also depends on how the resulting classes distinguish among the available  
592 irrigation fractions.

593 Both storage strata selected the same fraction for a given rainfall group at the 95% and 98% grain targets. Storage–  
594 rainfall and matched rainfall-only strategies consequently produced identical complete histories of irrigation, crop  
595 states, grain, ET and drainage. The storage split supplied additional environmental description without changing an  
596 irrigation action in this five-level quota design. All nine later years with eligible observations fell in the lower-  
597 storage category, so the higher-storage choices also lacked later-period exposure (Table 5; Figure 8). These features  
598 delimit the configuration and conditions represented by the equal outcomes.

599 Monitoring coverage must remain distinct from the grouping criterion. Rainfall-available controls can change  
600 irrigation during GRACE gaps, and those earlier decisions alter the soil-water state inherited in later years. Their  
601 outcomes therefore combine rainfall-based selection with additional years of management. Matched coverage  
602 preserves the appropriate comparison for the incremental effect of storage information. Finer quota choices,  
603 alternative thresholds and verified release availability would define different decision experiments with their own  
604 selection and evaluation requirements.

#### 605 **4.5. Crop-response constraints and management reliability**

606 Management selection depends on the modeled marginal response to water, in addition to absolute grain and canopy  
607 agreement. Intercepted-PAR radiation-use efficiency (Kiniry et al., 1989), cohort-specific leaf area and thermal  
608 senescence connect canopy development with carbon supply; grain number and fixed individual grain mass  
609 constrain the sink. Root uptake and the assimilation stress exponent then determine how water shortage changes

610 production. Variation among wheat-model transpiration formulations demonstrates the sensitivity of water-use  
611 predictions to these processes (Cammarano et al., 2016). Water coefficients near their prescribed bounds remain  
612 effective empirical estimates whose physiological interpretation and uniqueness are limited.

613 The estimation and evaluation data provide different evidence for these responses. Earlier growth estimates use 480  
614 wheat and 433 maize observations across four wheat sites and five maize sites, with whole site-years and balanced  
615 site contributions. Incomplete station irrigation histories limit transfer comparisons, and station observations have  
616 no weight in the fitted water-response objective. Wuqiao 2016–2018 supplies the documented treatment seasons for  
617 that fit. Wuqiao 2019 supplies a single test year with reported total storage, assumed layer distributions and  
618 transferred soil hydraulics; prior inspection of its outcomes makes the test retrospective (Figures 3–5). Regional  
619 comparison years also overlap crop-fitting years, so regional production differences characterize model scenarios  
620 rather than time-forward crop validation.

621 The absolute regional production reference also depends on the prescribed management and crop parameters.  
622 Conventional simulations produce  $8.065 \text{ t ha}^{-1}$  of wheat dry grain and  $8.691 \text{ t ha}^{-1}$  of maize dry grain per year.  
623 These values represent fixed irrigation dates, crop calendars and nutrition modifiers rather than an independently  
624 validated regional harvest inventory. The  $0.732 \text{ t ha}^{-1} \text{ yr}^{-1}$  allocation advantage depends on the same simulated  
625 marginal responses; its regional scaling does not remove field-response uncertainty.

626 Substantial field-test errors constrain confidence in irrigation-response transfer. Wheat grain nRMSE was 83.27%,  
627 with a bias of  $+3.108 \text{ t ha}^{-1}$  at the published 13% moisture basis; the absolute W3–W0 ET-contrast error was 116.37  
628 mm. Maize ET NSE was negative, with an ET-contrast error of 52.74 mm (Table 3; Table S5). These discrepancies  
629 affect the response functions used to select regional allocations. Numerical water-balance closure demonstrates  
630 conservation but supplies no measure of response accuracy. The eight purposive actual-cell checks address a  
631 separate uncertainty, spatial aggregation, and provide no regional field-validation sample. Depth-resolved soil water  
632 and complete dated management would better constrain uptake and redistribution; canopy and soil-moisture  
633 assimilation also requires treatment of retrieval errors and observation operators (Ines et al., 2013).

634 Production-retention targets introduce a further management distinction between period means and annual  
635 reliability. Mean retention of 96.27% and 99.63% coincided with annual minima of 92.12% and 98.58%; the 95%  
636 strategy met that threshold in eight of nine class-available years (Figure 8). A mean target consequently supplies no  
637 shortfall probability or guarantee for an individual farm. Local production floors, acceptable shortfall frequency and

638 a reserve entering the next crop can complement regional delivery and consumption constraints. Forecast-based  
639 irrigation studies illustrate explicit treatment of water-stress risk (Roy et al., 2021). Field scheduling additionally  
640 requires current soil-water information, weather forecasts and application timing.

#### 641 **4.6. Limitations and future directions**

642 The principal limitation is uncertainty in the crop responses used for strategy selection. Field testing covers a single,  
643 previously inspected year, while incomplete station management and assumed initial water profiles constrain  
644 transfer. Remaining errors in ET and grain responses prevent the simulated policy advantages from being  
645 interpreted as independently validated field benefits. Further evaluation requires new site-years with complete  
646 management records, withheld from both model development and strategy selection. Concurrent measurements of  
647 soil water by depth, ET, canopy development and harvest would allow separate assessment of water redistribution,  
648 crop stress and irrigation-treatment contrasts.

649 Regional results also depend on a static rotation map, representative climate–soil units, prescribed irrigation dates  
650 and fixed nutrition modifiers. Checks at purposively selected source cells cannot bound regional uncertainty.  
651 Dynamic nitrogen cycling is absent, and changes in crop area or actual farm practices are not reconstructed. Future  
652 analyses should expand evaluation at individual source cells and propagate uncertainty in crop parameters, soil  
653 hydraulics, weather and spatial aggregation through paired policy histories. Allocation stability and the magnitude  
654 and persistence of local losses require uncertainty assessment alongside the regional mean.

655 The storage comparison is specific to the tested quota grid, classification thresholds and lower-storage conditions  
656 represented during later comparison years. GRACE release availability before sowing remains unverified, so the  
657 scenarios do not establish operational scheduling performance. Prospective decision experiments should use  
658 information available on each decision date, maintain matched rainfall-only controls and evaluate both storage  
659 states. Finer quotas and application timing informed by soil water could be tested with constraints on consumptive  
660 use, local production and the risk of production shortfalls.

661 Finally, the profile uses free drainage at its lower boundary, with no capillary supply or deeper subsurface storage,  
662 and field irrigation is not linked to measured abstraction. Groundwater applications require joint accounting of soil  
663 water and aquifer flows, constrained by observations of pumping, recharge and storage change. Delivery capacity,  
664 irrigation costs and farmer participation also require assessment before the agronomic scenarios can support  
665 operational management.

## 666 **5. Conclusions**

667 At an identical 50% irrigation reduction, targeted allocation retained 0.732 t ha<sup>-1</sup> yr<sup>-1</sup> more simulated combined  
668 wheat–maize dry grain than uniform cuts during 2014–2025. This reduced regional grain loss by 7.219 million t  
669 yr<sup>-1</sup> (4.87% of uniform production), while increasing crop-plus-fallow ET by 20.76 mm yr<sup>-1</sup> (2.048 km<sup>3</sup> yr<sup>-1</sup>).  
670 Relative to uniform allocation, mean grain losses occurred on 37.60% of mapped rotation area and persisted in all  
671 twelve years on 21.34%. Regional production retention therefore requires separate consumption and local  
672 production constraints.

673 Annual strategies selected at 95% and 98% grain-retention targets retained 96.27% and 99.63% of conventional  
674 production over nine years with eligible GRACE observations. The 95% strategy reduced irrigation by 130.72 mm  
675 and ET by 36.84 mm per rotation. Matched rainfall-only controls selected identical quotas and produced identical  
676 outcomes, so storage classification provided no additional decision benefit within the tested quota design and  
677 monitoring coverage.

678 Continuous rotation accounting preserves the influence of preceding management on subsequent production and  
679 water use. The field-test errors and incomplete station management delimit confidence in transferring the estimated  
680 crop responses. Translation of these regional scenarios into irrigation management depends on observed response  
681 accuracy, explicit production-risk constraints and field scheduling informed by soil water and forecasts.  
682 Groundwater effects additionally require pumping, recharge and deeper subsurface storage beyond the crop-model  
683 boundary.

## 684 **Data and code availability**

685 Model code, frozen calibration and regional inputs, observation subsets and quality exclusions, weather, calibrated  
686 parameters, predictions, evaluation metrics and figure-data tables are available at [https://github.com/SmartAG-](https://github.com/SmartAG-Team/agri_water_management)  
687 [Team/agri\\_water\\_management](https://github.com/SmartAG-Team/agri_water_management). The repository includes a standalone Open Crop Model package, pinned  
688 dependencies, reproduction commands and SHA-256 provenance. Reproduction uses the included inputs without  
689 another model checkout or access to the historical workspace. Original datasets retain their provider licences and  
690 attribution requirements, documented in the repository source catalogue.

## 691 **Declaration of generative AI and AI-assisted technologies**

692 AI assisted with grammar checking.

## 693 **Supplementary material**

## 694 **Supplementary methods and results**

### 695 **S1. Observation sources and eligibility**

696 The station archive preserves 132 crop–plot–season trajectories, whole site-year partitions and source metadata. The  
697 7,397 eligible records occur in 127 cases and retain all five observation sites for their available variables.

698 Calibration and retrospective-evaluation trajectory counts remain 71 and 61. Variable-specific availability and  
699 missing combinations are retained in the accompanying case and site tables; missing measurements do not become  
700 zero observations.

701 Wheat LAI above  $7 \text{ m}^2 \text{ m}^{-2}$  remains excluded under the study-specific rule. Duplicated six-replicate maize harvest  
702 vectors in Yucheng 2012 and 2017 account for 24 excluded grain and biomass records, divided equally between the  
703 original calibration and evaluation partitions. These source-based exclusions retain raw values and do not depend  
704 on prediction residuals.

705 Earlier multisite growth estimates retain 480 wheat and 433 maize calibration observations. Current station growth  
706 and ET comparisons are conditional on archived management whose full irrigation-log completeness is unresolved.  
707 They have zero weight in the water-parameter fit. Wuqiao 2016–2018 supplies 12 documented treatment seasons  
708 per crop for that fit (Yang et al., 2024); 2019 supplies four retrospective testing seasons per crop. Seasonal ET, grain  
709 and biomass targets refer to the same treatment-season population, while ET contrasts are derived from those  
710 seasonal targets.

711 Twenty-six Fengqiu cases retain antecedent moisture initialization from the preceding calendar month. Complete  
712 fifteen-layer 0–1.5 m volumetric profiles map conservatively to the represented model layers, preserving depth-  
713 integrated storage. The identical eligibility rule applies to 11 original calibration and 15 evaluation cases. Monthly  
714 timing, aggregation and subplot support remain unresolved relative to the exact sowing-day state. Original cases  
715 remain available when a measured antecedent profile is absent.

716 **Table S1. Crop seasons and calibration/evaluation case counts by site.**

| Site      | Crop  | Years                                                                                                                                                         | Calibration cases | Evaluation cases |
|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------------------|
| Fengqiu   | Maize | 2004, 2005, 2006,<br>2007, 2008                                                                                                                               | 11                | 9                |
| Fengqiu   | Wheat | 2004, 2005, 2006,<br>2007, 2008                                                                                                                               | 11                | 10               |
| Gucheng   | Maize | 2018, 2019, 2020                                                                                                                                              | 8                 | 4                |
| Luancheng | Maize | 2004, 2005, 2006,<br>2007, 2008                                                                                                                               | 2                 | 3                |
| Luancheng | Wheat | 2004, 2005, 2006,<br>2007, 2008                                                                                                                               | 2                 | 3                |
| Shangqiu  | Maize | 2002, 2004, 2005,<br>2006, 2007                                                                                                                               | 2                 | 6                |
| Shangqiu  | Wheat | 2001, 2002, 2003,<br>2004, 2005, 2006,<br>2007                                                                                                                | 9                 | 8                |
| Yucheng   | Maize | 1998, 2000, 2001,<br>2002, 2004, 2005,<br>2006, 2007, 2008,<br>2009, 2010, 2011,<br>2012, 2013, 2014,<br>2015, 2016, 2017,<br>2018, 2019, 2020,<br>2021, 2022 | 14                | 9                |
| Yucheng   | Wheat | 1999, 2000, 2004,<br>2005, 2006, 2007,<br>2008, 2009, 2010,<br>2011, 2012, 2013,<br>2014, 2015, 2016,<br>2017, 2018, 2019,<br>2020, 2021, 2022                | 12                | 9                |

717 Years denote archived crop-season groups, which retain all their plot cases. Coverage is intermittent; years absent  
718 from the table are not assumed to have observations.

719 **Table S2. Source datasets for evaluated field observations.**

| Site      | Dataset         | DOI                       | Source                |
|-----------|-----------------|---------------------------|-----------------------|
| Fengqiu   | 2004-2008年河南封丘站 | 10.12199/nesdc.ecodb.mon. | National Ecosystem    |
|           | 作物生育及生长动态监测     | 2020.dp2011.fqa.002       | Science Data Center / |
|           | 数据              |                           | Science Data Bank     |
| Fengqiu   | 2005-2008年河南封丘站 | 10.12199/nesdc.ecodb.mon. | National Ecosystem    |
|           | 农田土壤水分物理性质监     | 2020.dp2011.fqa.006       | Science Data Center / |
|           | 测数据             |                           | Science Data Bank     |
| Gucheng   | 2018-2021年华北北部不 | 10.57760/sciencedb.o00119 | National Ecosystem    |
|           | 同播期对玉米生长发育的     | .00017                    | Science Data Center / |
|           | 影响数据集           |                           | Science Data Bank     |
| Luancheng | 2000-2008年河北栾城站 | 10.12199/nesdc.ecodb.mon. | National Ecosystem    |
|           | 作物生育及生长动态监测     | 2020.dp2011.lca.002       | Science Data Center / |
|           | 数据              |                           | Science Data Bank     |
| Shangqiu  | 2000-2007年河南商丘站 | 10.12199/nesdc.ecodb.mon. | National Ecosystem    |
|           | 作物生育及生长动态监测     | 2020.dp2011.sqa.003       | Science Data Center / |
|           | 数据              |                           | Science Data Bank     |
| Yucheng   | 1998-2006年山东禹城站 | 10.12199/nesdc.ecodb.mon. | National Ecosystem    |
|           | 作物生育及生长动态监测     | 2020.dp2011.yca.003       | Science Data Center / |
|           | 数据              |                           | Science Data Bank     |
| Yucheng   | 禹城站2005-2022年生物 | 10.12199/nesdc.ecodb.2021 | National Ecosystem    |
|           | 要素观测数据集         | YFF0703900.yca.2025.2     | Science Data Center / |
|           |                 |                           | Science Data Bank     |
| Yucheng   | 禹城站2005-2022年水环 | 10.12199/nesdc.ecodb.2021 | National Ecosystem    |
|           | 境要素观测数据集        | YFF0703900.yca.2025.3     | Science Data Center / |
|           |                 |                           | Science Data Bank     |

720 Dataset DOIs identify official source records; original metadata and observation-row provenance are retained in the  
721 reproducibility archive.

722 **S2. Crop processes, parameter definitions and water accounting**

723 **S2.1. Model structure and phenological development**

Open Crop Model simulates phenology, green leaf area, live and dead aboveground biomass, grain dry matter, rooting depth and layered soil-water storage. Inputs comprise weather, soil profiles, crop parameters, dated sowing and harvest, and irrigation events. Soil water carries continuously between crop and fallow segments in regional histories; Wuqiao crop starts retain their separately specified storage conditioning.

Wheat stage development combines temperature, vernalization and photoperiod modifiers with inherited stage requirements. Maize uses a subdaily temperature response reconstructed from daily extrema, with base, optimum and maximum temperatures of 6, 30 and 44 °C. The shared medium-maturity maize card requires approximately 1,909 °C d to reach R6. DVS anchors are wheat BBCH 09, 61 and 89 and maize VE, R1 and R6 (Meier, 2018).

Wheat dough stage and full ripeness retain their distinct definitions.

Seed-layer water above wilting-point storage permits germination. The represented layer is selected by sowing depth, with a 50-mm prior when no depth is supplied. Thermal development and wheat vernalization begin from the first eligible germination day rather than accumulating during dry-seed days. This layer-average availability is an effective establishment condition, not a measured seed-scale water potential. Fixed stage coefficients therefore do not imply fixed calendar-stage predictions across water states. New biomass production ends at wheat full ripeness or maize physiological maturity; green maize area can persist under the terminal senescence rate until harvest. Harvest is a separate management endpoint and does not substitute for an observed maturity date.

## **S2.2. Canopy development, biomass accumulation and grain yield**

Green cover is  $f_c = 1 - \exp(-k_L \text{LAI})$ . Daily aboveground dry-matter production is  $\Delta B = 10R_s p_{\text{PAR}} f_c \varepsilon f_T f_W^y f_N$ , where  $R_s$  is incident shortwave radiation,  $p_{\text{PAR}} = 0.45$ ,  $\varepsilon$  is RUE per MJ of intercepted PAR,  $f_T$  is the temperature modifier,  $f_N$  is the prescribed nutrition factor and  $f_W$  is realized transpiration divided by remaining potential transpiration demand after wet-canopy evaporation. The factor 10 converts  $\text{g m}^{-2}$  to  $\text{kg ha}^{-1}$ . PAR enters once (Kiniry et al., 1989). The fitted empirical assimilation exponents are 1.85985 for wheat and 0.25000 for maize. They govern carbon response separately from leaf-expansion and leaf-survival sensitivity.

SLA is linearly interpolated through DVS 0, 0.5, 1 and 2, with the DVS-2 value equal to DVS 1. New leaf dry matter acquires area using current SLA; established cohorts retain their birth area-to-mass ratio (de Wit and Berghuijs, n.d.). Thermal senescence hazard  $h(\text{DVS})$  is interpolated through DVS 0, 1, 1.4, 1.7 and 2, with a constant preflowering rate and increasing grain-filling rates (Tan et al., 2020). Daily survival

752 is  $\exp[-h(\text{DVS}) \max(T_{\text{mean}} - T_{\text{base}}, 0)](1 - q_s)$ , with  $q_s = 0.025(1 - f_w f_N)$ . The terminal hazard  
 753 continues at maize R6, whereas wheat BBCH 89 senesces remaining green leaves. LAI emerges from the  
 754 cohort area balance; no maximum-LAI parameter or cap enters this formulation.

755 Leaf allocation follows crop development, with separate early and late vegetative multipliers for maize and  
 756 additional temperature, water-supply and nutrition limits on leaf expansion. Initial LAI is an emergence condition.  
 757 Stage-specific SLA and senescence coefficients retain their inherited estimates and conditional trait priors (APSIM  
 758 Initiative, 2026; de Wit, 2022; de Wit, 2020); their values do not represent universal physiological bounds.

759 Grain number is fixed at the first grain-set day as  $N_g = 0.1B_f c_g$ , where  $B_f$  is simulated aboveground dry  
 760 mass immediately before grain set and  $c_g$  is the grain-number coefficient. Grain set starts at wheat BBCH  
 761 61 and maize R1. Maximum grain dry mass is  $Q_g = 10N_g m_g$ , with fixed  $m_g$  of 0.045 g grain<sup>-1</sup> for wheat  
 762 and 0.300 g grain<sup>-1</sup> for maize (APSIM Initiative, 2026). Assimilate supply and development-dependent  
 763 filling capacity constrain grain accumulation. Wheat remobilization transfers an explicit subset of stem  
 764 mass to grain while conserving total aboveground mass. Live and dead leaves, stems and grain remain in  
 765 the aboveground carbon pool; observed yield or harvest index does not initialize grain.

766 Prescribed nutrition factors distinguish archived fertilized and unfertilized management classes. Wuqiao and  
 767 regional simulations retain the fertilized class. These factors describe effective growth limitations; the model does  
 768 not resolve a mechanistic nitrogen balance.

### 769 **S2.3. Soil-water balance and actual evapotranspiration**

770 Potential transpiration and exposed-soil evaporation are  $T_p = K_T \text{ET}_0 f_c$  and  $E_p = K_E \text{ET}_0 (1 - f_c)$ , with  
 771  $\text{ET}_0$  supplied by AgERA5. Rainfall interception fills a canopy store of capacity 1 mm  $\times f_c$ . Wet-canopy  
 772 evaporation reduces the transpiration demand remaining within each outer water step. Actual ET is  
 773  $T_a + E_s + E_c$ , the sum of transpiration, soil evaporation and canopy evaporation. Wuqiao residue cover is  
 774 unmeasured and its exposed-soil boundary is retained; prescribed organic mulch, where supplied, reduces  
 775 potential soil evaporation using the FAO-56 cover approximation (Allen et al., 1998).

776 Two-stage evaporation depends on actual surface-layer storage. Total evaporable water is  
 777  $\text{TEW} = (\theta_{\text{FC}} - \theta_{\text{AD}})\Delta z_1$  and readily evaporable water is  $\text{REW} = 0.30\text{TEW}$ . Surface depletion is  
 778  $D_e = (\theta_{\text{FC}} - \theta_1)\Delta z_1$ . The evaporation reduction is  $K_r = \min \left[ 1, \max \left( 0, \frac{\text{TEW} - D_e}{\text{TEW} - \text{REW}} \right) \right]$ , with actual

779 extraction bounded by storage above air dry. Ponded water evaporates before soil storage. This storage  
 780 closure follows the two-stage drying principle (Allen et al., 1998) without resolving a complete dual-crop-  
 781 coefficient surface balance.

782 Root activity integrates an exponential depth distribution over rooted parts of each layer and is normalized  
 783 over the represented rooted volume. Local depletion stress is  $s_i = \min \left[ 1, \frac{\max(0, W_i - \theta_{WP,i} \Delta z_i)}{(1-p)(\theta_{FC,i} - \theta_{WP,i}) \Delta z_i} \right]$ , where  $p$   
 784 is the fitted readily available water fraction. Base uptake combines root activity, local stress, atmospheric  
 785 demand and the daily extraction coefficient; accessible storage and extraction capacity bound each  
 786 contribution. The compensation fraction allocates a bounded part of unmet demand to rooted layers whose  
 787 stress factors equal or exceed the activity-weighted mean. Total uptake remains bounded by demand times  
 788 the best rooted-layer stress factor and by available extraction capacity. Compensation redistributes  
 789 demand and does not eliminate uniformly dry-profile stress.

790 Effective saturation is  $S_e = \frac{\theta - \theta_{AD}}{\theta_{SAT} - \theta_{AD}}$ . With  $m = 1 - \frac{1}{n}$ , the retention relation is  $S_e = [1 + (\alpha | h | )^n]^{-m}$   
 791 (van Genuchten, 1980), and conductivity is  $K = K_{sat} S_e^{1/2} [1 - (1 - S_e^{1/m})^m]^2$  (Mualem, 1976; van  
 792 Genuchten, 1980). Adjacent half-layer saturated resistances define  $K_{sat, face} = \frac{d}{\frac{\Delta z_i}{2K_{sat,i}} + \frac{\Delta z_{i+1}}{2K_{sat,i+1}}}$ , where  
 793  $d = \frac{\Delta z_i + \Delta z_{i+1}}{2}$ . Relative-arithmetic face mobility is  $r_{face} = \frac{\Delta z_{i+1} r_i + \Delta z_i r_{i+1}}{\Delta z_i + \Delta z_{i+1}}$ , with  $r_i = \frac{K_i}{K_{sat,i}}$ . Thus  
 794  $K_{face} = K_{sat, face} r_{face}$ . This interface approximation permits a wet layer to supply a dry neighbor while  
 795 retaining the resistance of low-conductivity material.

796 Signed interface flux is  $q_i = K_{face} \left[ 1 + \frac{h_i - h_{i+1}}{d} \right]$ , positive downward. Transfers have equal and opposite  
 797 contributions in neighboring layer balances and are bounded by donor storage and receiving pore capacity.  
 798 Twenty-four outer steps partition daily forcing, evaporation and uptake; adaptive internal transport steps  
 799 limit net  $\theta$  increments to 0.005. The free-drainage bottom flux equals the current unsaturated conductivity  
 800 under unit hydraulic gradient. Upward redistribution is an internal profile flux, with no external capillary-  
 801 rise supply. This bounded explicit layered approximation does not resolve the full implicit Richards  
 802 system or saturated positive pressure.

803 For a field of profile storage  $S$ , the accounting identity is  $\Delta S = P + I - ET_a - D - Q$ . Applied irrigation  $I$   
 804 is a field-boundary input. Flood-irrigation scenarios wet the entire field, with application, drift and  
 805 conveyance losses set to zero at that boundary. Drainage  $D$  leaves the modeled profile and is distinct from

806 water reaching an aquifer after deeper vadose-zone transport (Wu et al., 2023).

## 807 **S2.4. Hydraulic derivation, shared crop parameters and numerical checks**

808 Saxton–Rawls equations use normalized sand and clay fractions and organic matter percent (Saxton and  
809 Rawls, 2006). SoilGrids carbon in  $\text{dg kg}^{-1}$  is converted to organic matter percent by  $\frac{1.724 \times \text{SOC}}{100}$ ; bulk  
810 density in  $\text{cg cm}^{-3}$  is divided by 100; coarse fragments in permille are divided by 1,000 (ISRIC — World  
811 Soil Information, 2020). Regional porosity uses  $1 - \frac{\text{BD}}{2.65}$ , with prescribed density and gravel adjustments.  
812 Air-dry water content retains its declared prior. Field-capacity, wilting-point and saturation ordering is  
813 checked before simulation; unavailable complete profiles remain excluded under the spatial-source rule.

814 The van Genuchten  $\alpha$  and  $n$  values are conditional hydraulic priors reconstructed from supplied field-  
815 capacity and wilting-point endpoints at pressure heads of  $-3,300$  and  $-150,000$  mm, with  $\theta_{\text{AD}}$  as residual  
816 water and  $\theta_{\text{SAT}}$  as saturated content. Endpoint reconstruction preserves those storage thresholds rather  
817 than fitting curves to crop outcomes. Neither depth-resolved Wuqiao retention measurements nor exact  
818 seasonal initial layer distributions are available. The separate Wuqiao particle-density prior is  $2.75 \text{ g cm}^{-3}$ .

819 Table S3 contains effective shared crop values after applying the frozen fit multipliers. Regional and Wuqiao inputs  
820 use identical shared cards within each crop. Maximum root depth is limited by the supplied profile depth. Stage-  
821 specific SLA and senescence, fixed grain mass and inherited growth coefficients remain distinct from the eight  
822 water coefficients fitted in the current run. Near-zero root-density decay yields approximately uniform activity over  
823 represented rooted volume; it does not identify an observed root-mass profile.

824 The 164 selected station and field trajectories conserve daily water and aboveground carbon, with maximum  
825 absolute residuals of  $1.00 \times 10^{-12}$  mm and  $6.71 \times 10^{-12}$  kg ha $^{-1}$ . Exact inputs, observation identities, frozen  
826 parameters, source snapshots and SHA-256 provenance accompany the predictions. Conservation establishes  
827 arithmetic consistency; hydraulic and crop-response accuracy depend on the conditional observations and  
828 retrospective field test.

829 Eight existing water-response coefficients per crop are estimated from documented Wuqiao management in 2016–  
830 2018: the transpiration multiplier, soil evaporation coefficient, readily available water fraction, assimilation water-  
831 stress exponent, root-density decay, maximum rooting depth,  $\log_{10}$  daily root extraction coefficient and root  
832 compensation fraction. RUE, leaf allocation and grain-number multipliers remain fixed. Each crop contributes 12  
833 seasonal ET, 12 dry-grain and 12 dry-biomass targets, together with three W3–W0 ET contrasts derived from the

834 same seasonal targets. The criterion retains weights of 0.325, 0.175, 0.15 and 0.10 for those components. ET  
835 residual scales are  $\max(50 \text{ mm}, 0.15 \times \text{observed ET})$ ; grain and biomass scales are  $\max(1,000 \text{ kg ha}^{-1}, 0.20 \times$   
836  $\text{observed dry mass})$ . Contrast residuals use 50 mm. Eight coefficient-prior terms have total weight 0.01, with  
837 inherited bounds, centres and scales.

838 Bounded least squares starts from the inherited water vector, with coefficients scaled to their prescribed intervals, a  
839 40-evaluation limit and convergence tolerances of  $8 \times 10^{-5}$ . Reference and refitted coefficients use the identical  
840 field-only criterion; the equal-crop mean calibration data loss is 0.592580 for the reference and 0.577934 for the  
841 selected refit. Selection excludes prior penalties from this comparison and precedes 2019 testing. Station  
842 observations, daily ET and phase ET with unresolved sampling windows have zero weight in the water-parameter  
843 fit. Source observations, weather, soil priors, initialization rules and native process equations remain fixed during  
844 this refit.

845 **Table S3. Shared crop parameters transferred to regional simulations.**

| Parameter                                     | Unit                               | Wheat                    | Maize                    |
|-----------------------------------------------|------------------------------------|--------------------------|--------------------------|
| Radiation-use efficiency                      | g MJ <sup>-1</sup> intercepted PAR | 3.082080                 | 3.379974                 |
| Specific leaf area at DVS 0                   | m <sup>2</sup> g <sup>-1</sup>     | 0.028482                 | 0.024045                 |
| Specific leaf area at DVS 0.5                 | m <sup>2</sup> g <sup>-1</sup>     | 0.027578                 | 0.029960                 |
| Specific leaf area at DVS 1                   | m <sup>2</sup> g <sup>-1</sup>     | 0.023462                 | 0.024000                 |
| Leaf senescence rate before flowering         | (°C d) <sup>-1</sup>               | 0.000536                 | 0.000010                 |
| Leaf senescence rate at DVS 1.4               | (°C d) <sup>-1</sup>               | 0.001907                 | 0.000023                 |
| Leaf senescence rate at DVS 1.7               | (°C d) <sup>-1</sup>               | 0.005327                 | 0.001650                 |
| Leaf area index at emergence                  | m <sup>2</sup> m <sup>-2</sup>     | 0.141900                 | 0.070844                 |
| Grain number per biomass at grain set         | grains g <sup>-1</sup>             | 26.272389                | 4.242740                 |
| Maximum individual grain mass                 | g grain <sup>-1</sup>              | 0.045000                 | 0.300000                 |
| Transpiration coefficient                     | dimensionless                      | 0.838813                 | 1.023273                 |
| Soil evaporation coefficient                  | dimensionless                      | 1.295161                 | 0.100000                 |
| Root water uptake coefficient                 | d <sup>-1</sup>                    | 0.042671                 | 0.033543                 |
| Leaf biomass partitioning multiplier          | dimensionless                      | 1.010797                 | —                        |
| Early vegetative leaf partitioning multiplier | dimensionless                      | —                        | 0.912989                 |
| Late vegetative leaf partitioning multiplier  | dimensionless                      | —                        | 1.779539                 |
| Stem reserve fraction                         | dimensionless                      | 0.270612                 | —                        |
| Stem reserve release fraction                 | dimensionless                      | 0.067334                 | —                        |
| Assimilation water-stress exponent            | dimensionless                      | 1.859851                 | 0.250000                 |
| Readily available water fraction              | dimensionless                      | 0.543149                 | 0.674994                 |
| Root density decay                            | m <sup>-1</sup>                    | 3.41 × 10 <sup>-10</sup> | 2.27 × 10 <sup>-13</sup> |
| Maximum rooting depth                         | mm                                 | 1888.654910              | 1699.999998              |
| Root compensation fraction                    | dimensionless                      | 1.000000                 | 6.62 × 10 <sup>-8</sup>  |
| Readily evaporable water fraction             | dimensionless                      | 0.300000                 | 0.300000                 |

846 Effective shared crop parameters from the selected current cards. RUE uses intercepted PAR with PAR fraction  
847 0.45. DVS-2 SLA equals its DVS-1 value; senescence is a thermal hazard with a separate daily stress loss. Wheat  
848 BBCH 89 terminates remaining green leaf area; maize retains its terminal hazard at R6. Grain mass is fixed. Water

849 coefficients are estimated from Wuqiao 2016–2018; growth coefficients retain inherited multisite estimates. Root  
850 depth is limited by profile depth. Dashes denote crop-specific processes. No maximum-LAI parameter is fitted.

### 851 **S3. Study-area and spatial aggregation checks**

852 The SRTM-derived terrain grid (NASA JPL, 2013) contains 720 eligible lowland components; the connected  
853 component retained by the operational definition covers 404,658 km<sup>2</sup>. Disconnected eligible terrain totals 6,465 km<sup>2</sup>  
854 and contains approximately 7,068 ha of CPM-corrected rotation area. The main definition reaches the southern and  
855 eastern bounding-box edges, so geographic sensitivity remains relevant to regional interpretation. Nine elevation–  
856 slope alternatives combine elevations 100, 200 and 300 m with slopes 1°, 2° and 3°. Seven annual mapped-area  
857 layers cover 2015–2021.

858 A common-extent comparison between ChinaCP plus CPM (Qiu et al., 2022) and the 2020 China Crop Dataset  
859 layer (Fu et al., 2026; Fu et al., 2025) yields 9.694 and 7.841 million ha, respectively; CCD2020 is 19.11% lower on  
860 that common support. This disagreement is map uncertainty rather than proof of either product’s accuracy. The  
861 modeled 2020 area is held fixed to isolate climate and irrigation effects. Historical crop expansion, rotation  
862 switching and double-counting of harvested crop area are not estimated.

863 Thirty-two actual-cell representatives use rotation-area weights and selection-period climate means. Eight  
864 purposively selected non-representative cells test approximation under all five continuous irrigation fractions. Each  
865 crop–fraction comparison retains 96 evaluation cell-years in 2014–2025. Absolute grain, ET and management-  
866 response differences compare actual-cell with representative-cell model predictions. The sample measures  
867 aggregation sensitivity rather than observed regional accuracy or a probabilistic error bound (Table S4; Figure S1).

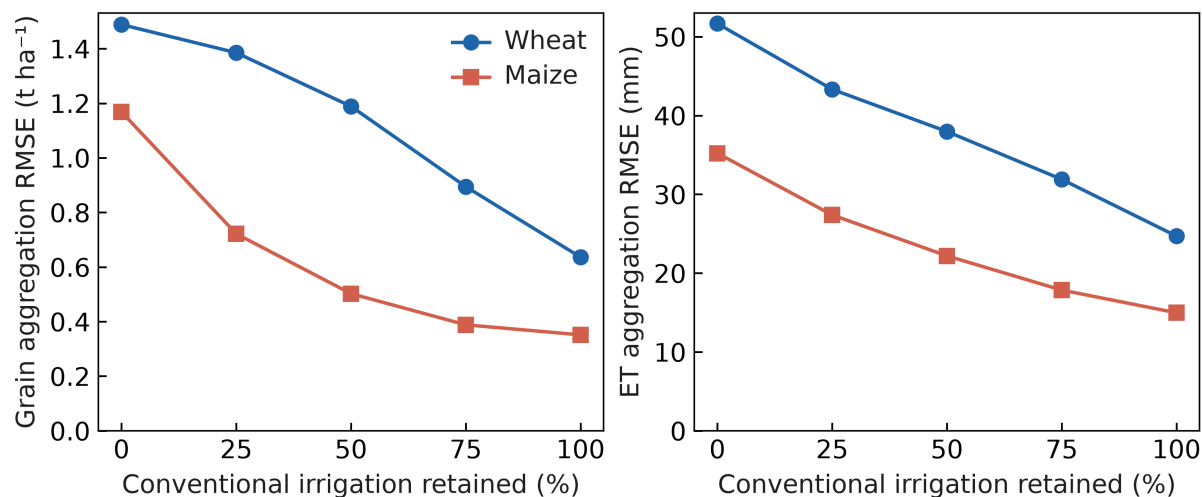
868 **Table S4. Crop-model spatial aggregation errors in eight purposively selected simulation cells.**

| Crop  | Irrigation<br>fraction | Variable    | Unit               | N  | RMSE  | Bias                     |
|-------|------------------------|-------------|--------------------|----|-------|--------------------------|
| Maize | 0.000                  | Grain yield | t ha <sup>-1</sup> | 96 | 1.17  | 0.18                     |
| Maize | 0.000                  | ET          | mm                 | 96 | 35.21 | 6.41                     |
| Maize | 0.250                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.72  | 0.13                     |
| Maize | 0.250                  | ET          | mm                 | 96 | 27.39 | 5.17                     |
| Maize | 0.500                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.50  | 0.08                     |
| Maize | 0.500                  | ET          | mm                 | 96 | 22.19 | 3.82                     |
| Maize | 0.750                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.39  | 0.08                     |
| Maize | 0.750                  | ET          | mm                 | 96 | 17.85 | 4.38                     |
| Maize | 1.000                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.35  | 0.09                     |
| Maize | 1.000                  | ET          | mm                 | 96 | 14.97 | 5.69                     |
| Wheat | 0.000                  | Grain yield | t ha <sup>-1</sup> | 96 | 1.49  | 0.19                     |
| Wheat | 0.000                  | ET          | mm                 | 96 | 51.66 | -3.18                    |
| Wheat | 0.250                  | Grain yield | t ha <sup>-1</sup> | 96 | 1.39  | 0.07                     |
| Wheat | 0.250                  | ET          | mm                 | 96 | 43.32 | -4.26                    |
| Wheat | 0.500                  | Grain yield | t ha <sup>-1</sup> | 96 | 1.19  | -7.41 × 10 <sup>-3</sup> |
| Wheat | 0.500                  | ET          | mm                 | 96 | 37.97 | -4.56                    |
| Wheat | 0.750                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.89  | -0.01                    |
| Wheat | 0.750                  | ET          | mm                 | 96 | 31.90 | -2.05                    |
| Wheat | 1.000                  | Grain yield | t ha <sup>-1</sup> | 96 | 0.64  | 0.11                     |
| Wheat | 1.000                  | ET          | mm                 | 96 | 24.70 | 3.21                     |

869 Current model aggregation discrepancies against eight actual-cell simulations under all five irrigation fractions.

870 Grain is converted from kg ha<sup>-1</sup> to t ha<sup>-1</sup> in every displayed grain-valued column. Exactly zero irrigation

871 discrepancies are omitted from display; full inputs and metrics remain in the current-results product.



872 **Figure S1. Current crop-model spatial aggregation errors under all five irrigation fractions in eight purposively selected cells during**  
873 **2014–2025. Actual-cell and representative predictions share the current native model and crop parameters. These discrepancies**  
874 **assess spatial representation rather than field-observation accuracy.**

#### 875 **S4. Annual field yields and irrigation-treatment contrasts**

876 Annual Wujiao grain targets comprise sixteen treatment-year means per crop in 2016–2019, digitized with standard  
877 errors from the primary Supplementary Figure S2. Grain retains its 13% moisture basis in comparisons. The twelve  
878 2016–2018 means per crop enter calibration after a single multiplication by 0.87 to dry matter. The four 2019  
879 means remain outside fitting and selection but had been inspected during earlier development. Four-year means  
880 summarize the same years and supply no additional independent targets.

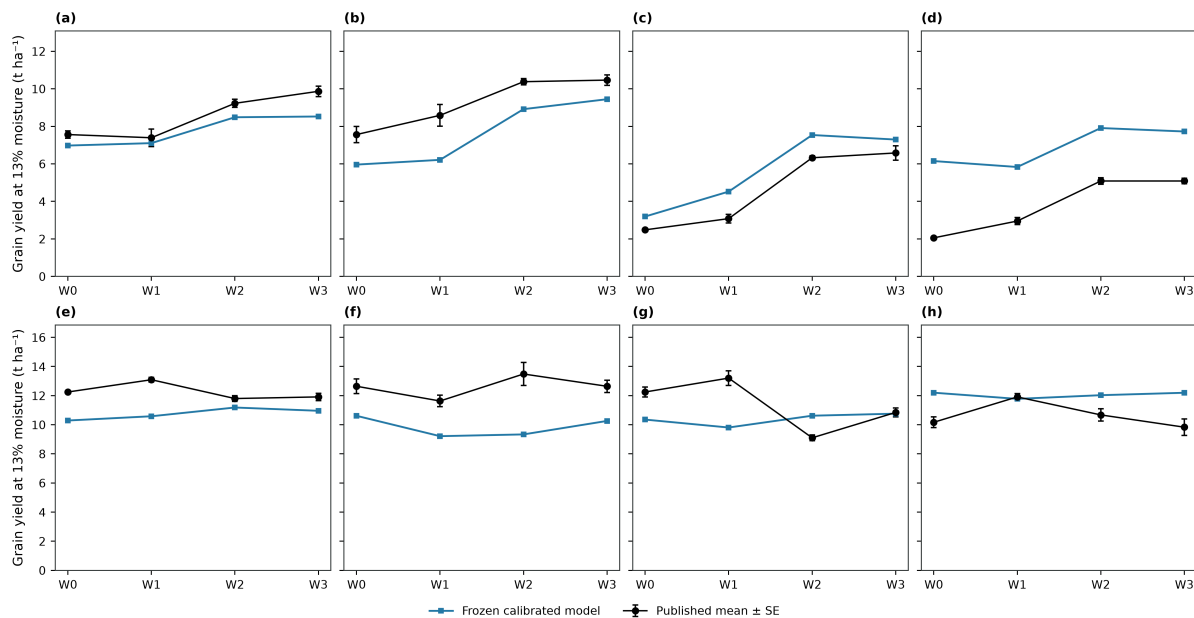
881 Annual grain calibration RMSE is 1.249 t ha<sup>-1</sup> for wheat and 2.267 t ha<sup>-1</sup> for maize; retrospective-testing RMSE is  
882 3.161 and 1.700 t ha<sup>-1</sup> at 13% moisture. Within-year W1–W3 minus W0 grain-contrast RMSE is 0.519 and 1.669 t  
883 ha<sup>-1</sup> in calibration and 1.316 and 1.323 t ha<sup>-1</sup> in 2019 (Table S5; Figure S2). Maize treatment labels retain  
884 preceding wheat management because maize irrigation is identical within each year.

885 All printed seasonal ET targets remain eligible after matching the 2016 maize rainfall period. Maize-season rainfall  
886 of 499.1 mm excludes the 100.5 mm during wheat from the 599.6-mm rotation total. Reconstructed balance  
887 residuals for W0–W3 are +0.1, +0.1, -0.9 and +0.1 mm. Source ET and depletion values remain unchanged.  
888 Storage-depletion diagnostics share the source ET balance and its unconfirmed exact sampling dates; they are  
889 conditional accounting comparisons rather than independent soil-water validation.

890 **Table S5. Wuqiao annual grain and within-year treatment contrasts by fitting and testing partition.**

| Crop  | Dataset                               | Comparison                       | N  | RMSE (t/ha) | Bias (t/ha) | NSE    | R <sup>2</sup> |
|-------|---------------------------------------|----------------------------------|----|-------------|-------------|--------|----------------|
| Maize | Calibration<br>(2016–2018)            | Annual grain<br>yield            | 12 | 2.267       | -1.738      | -2.940 | 0.128          |
| Maize | Calibration<br>(2016–2018)            | Within-year<br>W1–W3<br>minus W0 | 9  | 1.669       | 0.296       | -0.788 | 0.076          |
| Maize | Retrospective<br>evaluation<br>(2019) | Annual grain<br>yield            | 4  | 1.700       | 1.403       | -3.641 | 0.972          |
| Maize | Retrospective<br>evaluation<br>(2019) | Within-year<br>W1–W3<br>minus W0 | 3  | 1.323       | -0.836      | -1.403 | 1.000          |
| Wheat | Calibration<br>(2016–2018)            | Annual grain<br>yield            | 12 | 1.249       | -0.445      | 0.746  | 0.817          |
| Wheat | Calibration<br>(2016–2018)            | Within-year<br>W1–W3<br>minus W0 | 9  | 0.519       | 0.064       | 0.857  | 0.882          |
| Wheat | Retrospective<br>evaluation<br>(2019) | Annual grain<br>yield            | 4  | 3.161       | 3.108       | -4.654 | 0.870          |
| Wheat | Retrospective<br>evaluation<br>(2019) | Within-year<br>W1–W3<br>minus W0 | 3  | 1.316       | -1.312      | -0.712 | 0.994          |

891 Grain is at 13% moisture. Calibration comprises 2016–2018 and retrospective testing comprises 2019. Source  
892 means and standard errors are digitized from Supplementary Figure S2, with pixel tolerances near 0.043 t ha<sup>-1</sup> for  
893 wheat and 0.056 for maize. Contrasts compare W1–W3 with same-year W0. Four-year means overlap these annual  
894 targets; maize contrasts retain preceding wheat histories.



895 **Figure S2.** Annual Wuqiao grain at 13% moisture under the selected current parameters. Panels (a–d) show wheat and (e–h) maize;
 896 columns from left to right show 2016, 2017, 2018 and 2019. Black points and error bars are published means  $\pm$  standard error of
 897 three replicates; blue squares show current predictions. The 2016–2018 treatment means contribute to water calibration after
 898 conversion to dry grain; 2019 is retrospective testing, excluded from fitting and selection. Maize irrigation is identical within each
 899 year and labels retain preceding wheat management. Source means and SE are digitized from Supplementary Figure S2 of the
 900 primary experiment.

## 901 **S5. Class exposure, continuous states and rainfall-only controls**

902 All nine common GRACE-available years occupy lower-storage classes; three other years use conventional
 903 fallback. The selected current quotas are identical across storage strata within each rainfall group. Matched rainfall-
 904 only and storage–rainfall histories have identical actions, crop states and outcomes throughout the modeled record.
 905 Available-rainfall controls apply reduced quotas during monitoring gaps and therefore create different carried water
 906 states before later common years. These history effects are separate from the value of the storage classification.
 907 Table S7 reports all-year and common-year retention and water use; Table S8 reports the matched and available-
 908 rainfall controls.

909 **Table S6. Class exposure and missing-storage coverage by period.**

| Period     | Class                               | Unit-years | Years | Area-years (ha<br>yr) | Area-year<br>fraction |
|------------|-------------------------------------|------------|-------|-----------------------|-----------------------|
| evaluation | Lower storage /<br>lower rainfall   | 174        | 9     | 55410278.787          | 0.468                 |
| evaluation | Lower storage /<br>higher rainfall  | 114        | 9     | 33389136.355          | 0.282                 |
| evaluation | Missing storage                     | 96         | 3     | 29599805.048          | 0.250                 |
| selection  | Higher storage /<br>lower rainfall  | 68         | 6     | 19466918.675          | 0.179                 |
| selection  | Higher storage /<br>higher rainfall | 124        | 6     | 39732691.420          | 0.366                 |
| selection  | Lower storage /<br>lower rainfall   | 109        | 5     | 34747673.179          | 0.320                 |
| selection  | Lower storage /<br>higher rainfall  | 51         | 5     | 14585335.233          | 0.134                 |

910 **Table S7. Continuous class-adaptive irrigation evaluation during 2014–2025.**

| Strategy                 | Period              | Years | Grain retained<br>(%) | Irrigation cut<br>(mm) | ET cut (mm) | Minimum<br>grain (%) |
|--------------------------|---------------------|-------|-----------------------|------------------------|-------------|----------------------|
| Storage–<br>rainfall 95% | Nine GRACE<br>years | 9     | 96.270                | 130.721                | 36.842      | 92.115               |
| Storage–<br>rainfall 95% | All twelve<br>years | 12    | 96.818                | 98.040                 | 29.733      | 92.115               |
| Storage–<br>rainfall 98% | Nine GRACE<br>years | 9     | 99.626                | 35.721                 | 6.171       | 98.579               |
| Storage–<br>rainfall 98% | All twelve<br>years | 12    | 99.622                | 26.790                 | 5.208       | 98.579               |
| Conventional             | Nine GRACE<br>years | 9     | 100.000               | 0.000                  | 0.000       | 100.000              |
| Conventional             | All twelve<br>years | 12    | 100.000               | 0.000                  | 0.000       | 100.000              |

911 Current continuous histories. Grain retention is a ratio of period-mean regional dry-grain production to conventional  
 912 irrigation. ET includes crops and fallows. All-year summaries include conventional fallback; common-year  
 913 summaries use the nine GRACE-available years.

914 **Table S8. Continuous GRACE–rainfall and rainfall-only strategy comparison during 2014–2025.**

| Strategy                   | Period           | Grain retained (%) | Irrigation (mm) | ET (mm) |
|----------------------------|------------------|--------------------|-----------------|---------|
| Storage–rainfall 95%       | Nine GRACE years | 96.27              | 249.279         | 848.527 |
| Storage–rainfall 95%       | All twelve years | 96.82              | 281.960         | 857.698 |
| Storage–rainfall 98%       | Nine GRACE years | 99.63              | 344.279         | 879.197 |
| Storage–rainfall 98%       | All twelve years | 99.62              | 353.210         | 882.223 |
| Rainfall 95%,<br>available | Nine GRACE years | 95.40              | 249.279         | 844.203 |
| Rainfall 95%,<br>available | All twelve years | 95.04              | 248.565         | 842.945 |
| Rainfall 95%,<br>matched   | Nine GRACE years | 96.27              | 249.279         | 848.527 |
| Rainfall 95%,<br>matched   | All twelve years | 96.82              | 281.960         | 857.698 |
| Rainfall 98%,<br>available | Nine GRACE years | 99.41              | 344.279         | 878.151 |
| Rainfall 98%,<br>available | All twelve years | 99.36              | 343.565         | 879.669 |
| Rainfall 98%,<br>matched   | Nine GRACE years | 99.63              | 344.279         | 879.197 |
| Rainfall 98%,<br>matched   | All twelve years | 99.62              | 353.210         | 882.223 |

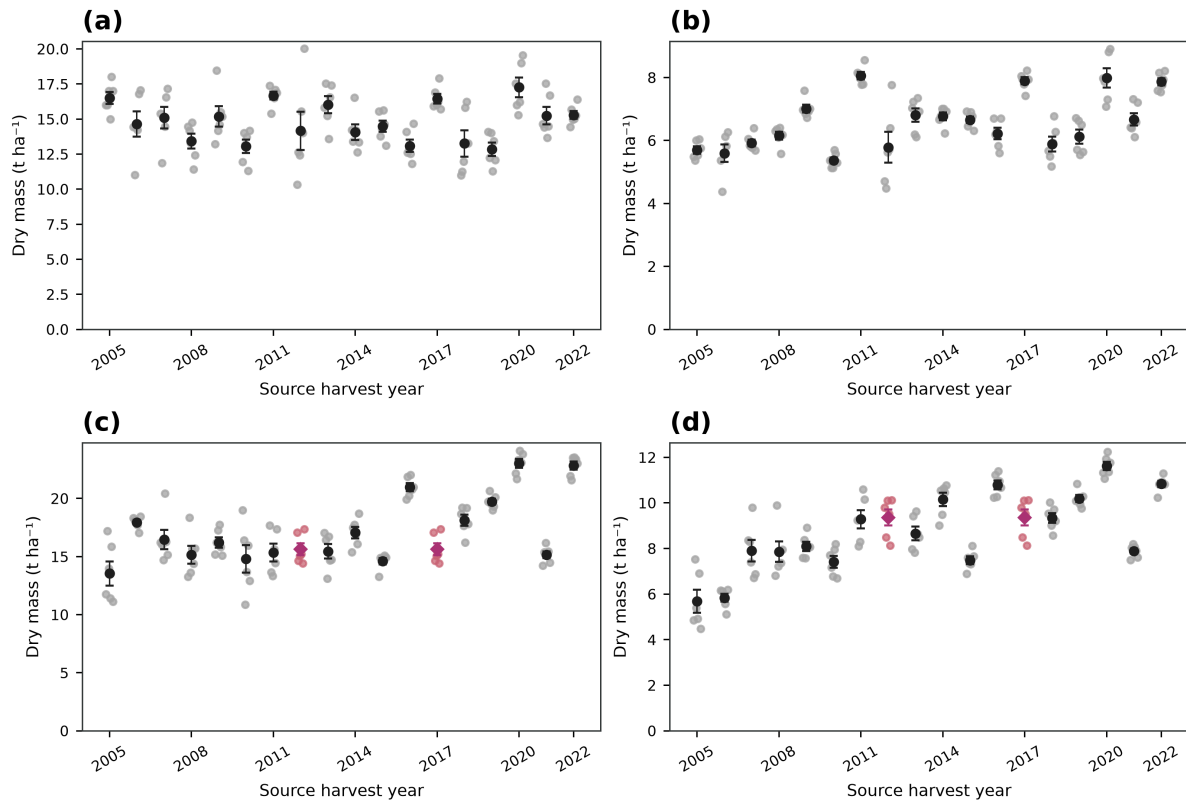
915 Matched controls share the complete GRACE-availability and fallback calendar. Available-rainfall controls act  
 916 during missing-storage years and alter subsequent soil states. All values use current parameters; differences do not  
 917 identify causal groundwater responses.



918 **Figure S3. Simulated 0–2 m soil-water storage in representative units (a) 0 and (b) 31. State carries continuously through wheat,**  
 919 **maize, fallows and annual irrigation choices. Displayed histories compare conventional irrigation with selected current strategies;**  
 920 **they are model states rather than observed groundwater levels.**

## 921 **S6. Harvest data quality and crop model recalibration**

922 Original Yucheng harvest sources contain identical six-replicate maize vectors in 2012 and 2017 under different  
 923 date and cultivar labels. Both groups are retained as raw evidence and excluded from scored grain and harvest  
 924 biomass. Eligibility depends on source authenticity rather than model residuals. Current plots preserve observed  
 925 quadrat variation and the original whole site-year partitions (Figure S4).



926 **Figure S4. Original Yucheng harvest observations at YCAZH01ABC\_01. Panels (a, b) show wheat aboveground biomass and grain;**  
 927 **panels (c, d) show maize aboveground biomass and grain. Faint points are six quadrat observations; dark markers and error bars**  
 928 **are means  $\pm$  replicate standard error. Magenta diamonds identify the identical full maize source vectors in 2012 and 2017. Raw**  
 929 **values remain displayed for source inspection; both groups are excluded from comparisons with quality-controlled observations.**  
 930 **Error bars describe sampling variation, rather than model or policy uncertainty.**

## 931 **S7. Spatial decomposition under the common irrigation budget**

932 Frozen area shares weight complete current irrigation histories before spatial aggregation. Dry-grain differences in  
 933 kg ha<sup>-1</sup> multiply represented hectares and divide by 1,000 for tonnes; water depths multiply hectares by 10 for cubic  
 934 metres. Independent sums reproduce regional contrasts. Latitude bands describe the distribution of gains and losses,  
 935 while the 3,641 mapped cells inherit 32 simulated representative responses (Table S9).

936 **Table S9. Spatial decomposition at a common 50% regional irrigation reduction during 2014–2025.**

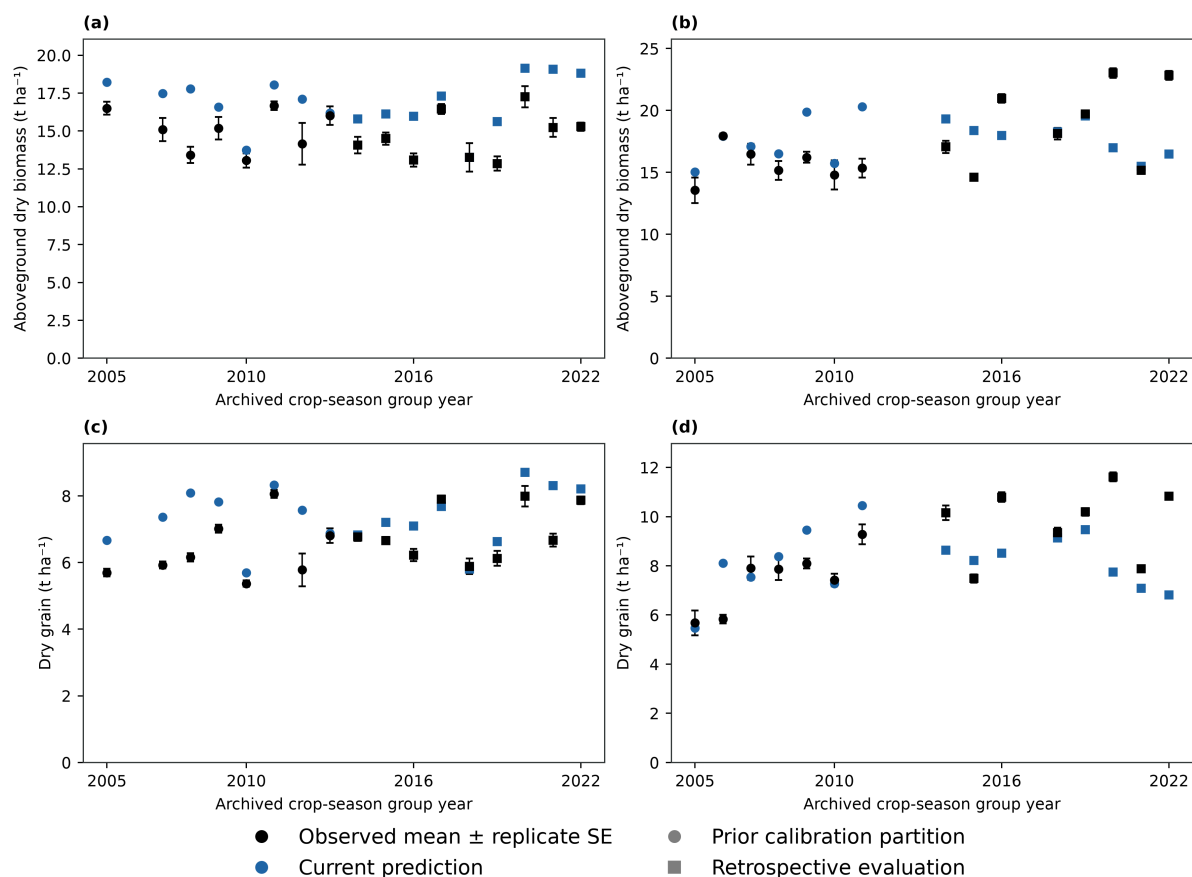
| Latitude band      | Area<br>share (%) | Targeted<br>irrigation<br>fraction | Grain change<br>(t ha <sup>-1</sup> ) | ET change (mm) | Grain contribution<br>(Mt) |
|--------------------|-------------------|------------------------------------|---------------------------------------|----------------|----------------------------|
| Southern (32–36°N) | 60.733            | 0.305                              | -0.357                                | -25.599        | -2.142                     |
| Central (36–38°N)  | 31.369            | 0.799                              | 2.415                                 | 93.497         | 7.475                      |
| Northern (38–41°N) | 7.897             | 0.813                              | 2.420                                 | 88.358         | 1.886                      |

937 Current targeted-minus-uniform means at the same 50% regional irrigation cut. Latitude bands are reporting  
938 partitions. Source cells inherit representative responses and are not independent observations.

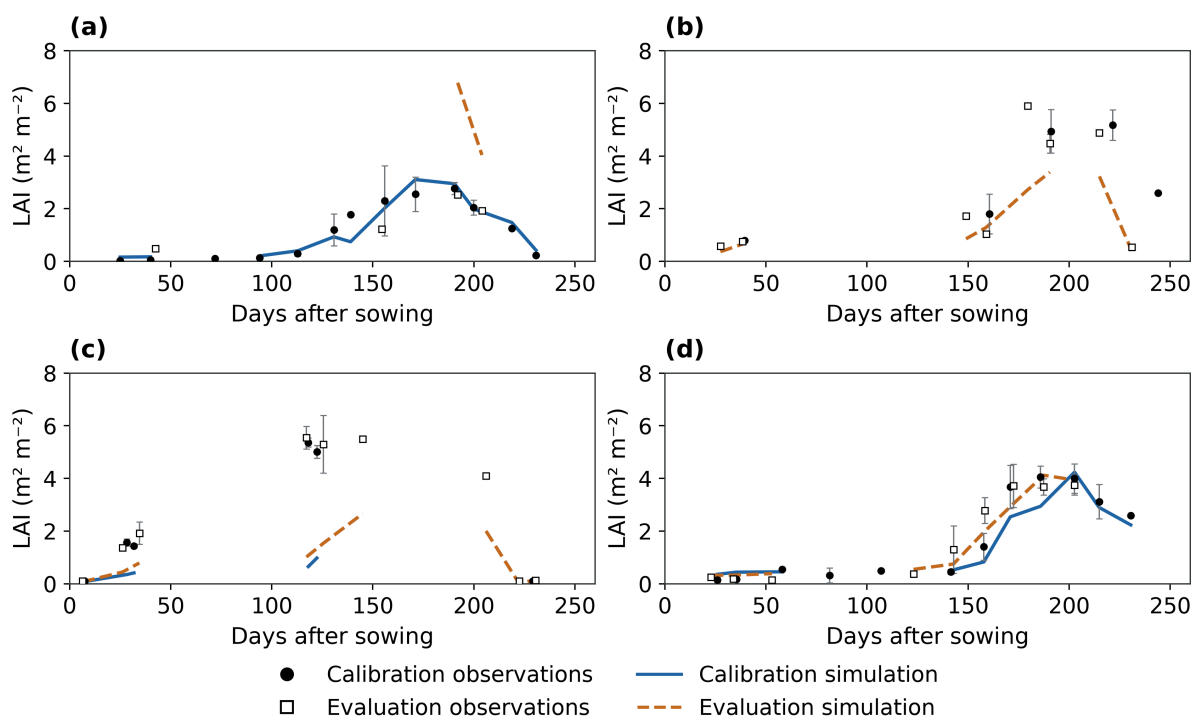
939 **S8. Seasonal crop trajectories and harvest outcomes**

940 Current seasonal curves use the selected frozen parameters and all sites with eligible observations for each variable.  
941 Observation and prediction channels share windows, case identities and aggregation weights. Bins span 15 days for  
942 wheat and 7 days for maize; site-year contributions and populations can change among bins. Standard errors  
943 describe variation across site-year means and are omitted for single-year bins. Missing bins remain gaps. Unbinned  
944 records determine performance scores (Figures S6–S11).

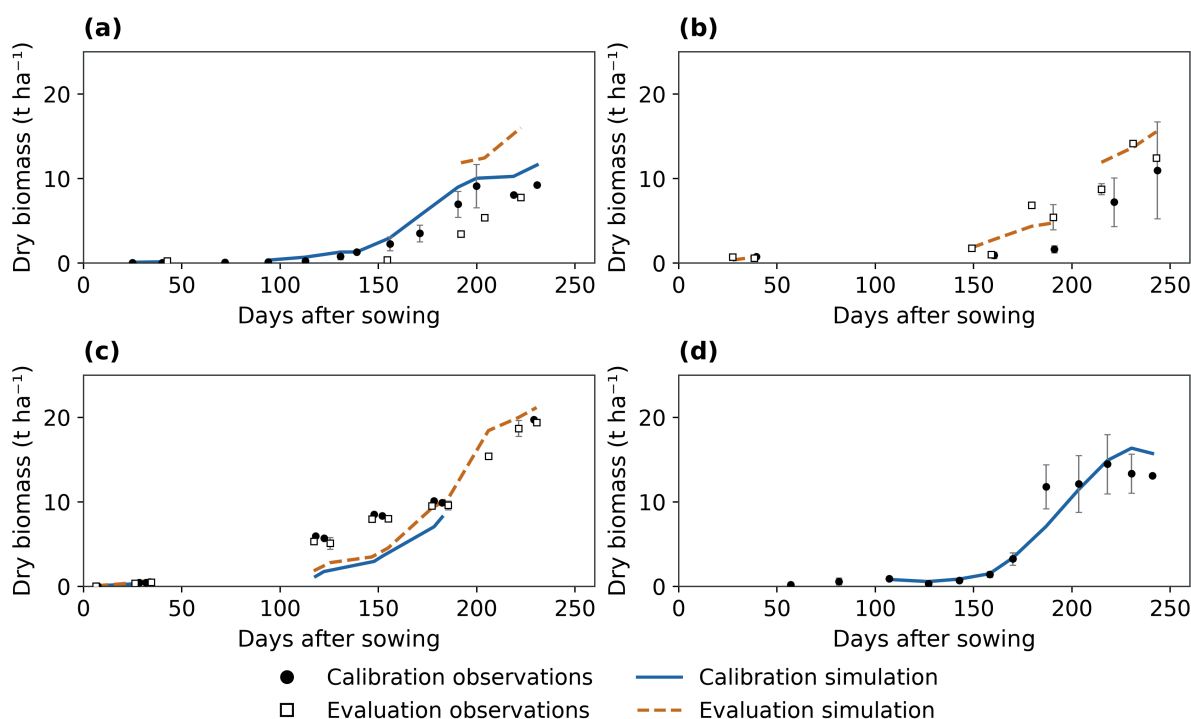
945 The original station calibration partitions retain earlier growth-fitting history; all station comparisons are  
946 conditional on unconfirmed complete irrigation management. Station observations have no weight in the latest  
947 water fit. Yucheng has no eligible in-season biomass evaluation observations, while its harvest-biomass endpoints  
948 remain separately available. Figure S5 shows dry-grain and aboveground harvest means with quadrat SE. Six  
949 samples within a crop-year describe sampling variation, and the duplicated 2012 and 2017 maize groups remain  
950 excluded. Exact case availability, site-specific metrics, bin contributions and harvest source rows accompany the  
951 figures.



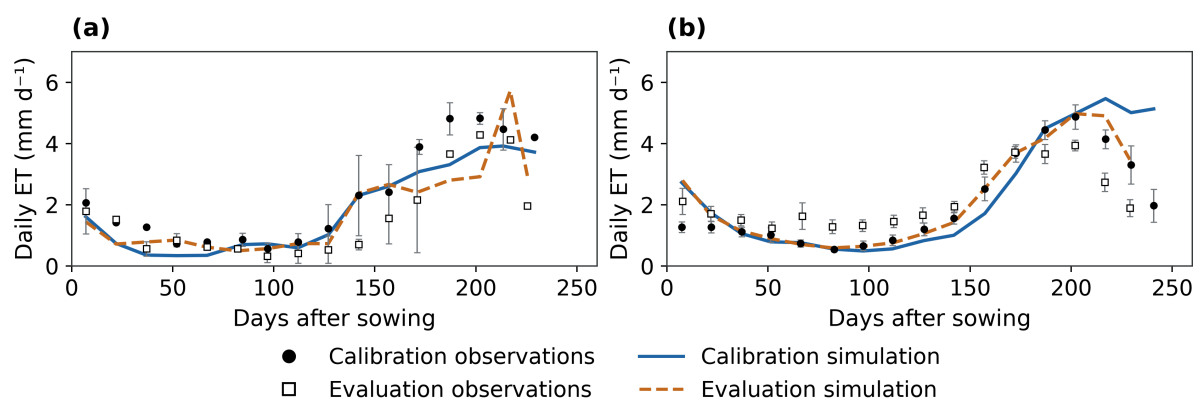
952 **Figure S5. Conditional Yucheng harvest comparisons. Wheat occupies the left column and maize the right; the top row shows**  
 953 **aboveground biomass and the bottom row grain. Black points show observed means  $\pm$  standard errors among six quadrat samples**  
 954 **per crop-year; blue points show current coupled-model predictions at the same sampling windows. Circles and squares distinguish**  
 955 **the original calibration and retrospective-evaluation partitions. Grain and total aboveground biomass are dry matter. The duplicated**  
 956 **maize 2012 and 2017 source vectors remain excluded. Within-year replicates describe spatial sampling variation and are not**  
 957 **independent crop seasons. Management completeness remains unconfirmed and current water coefficients use Wuqiao fitting targets,**  
 958 **while growth coefficients retain their earlier multisite fitting history.**



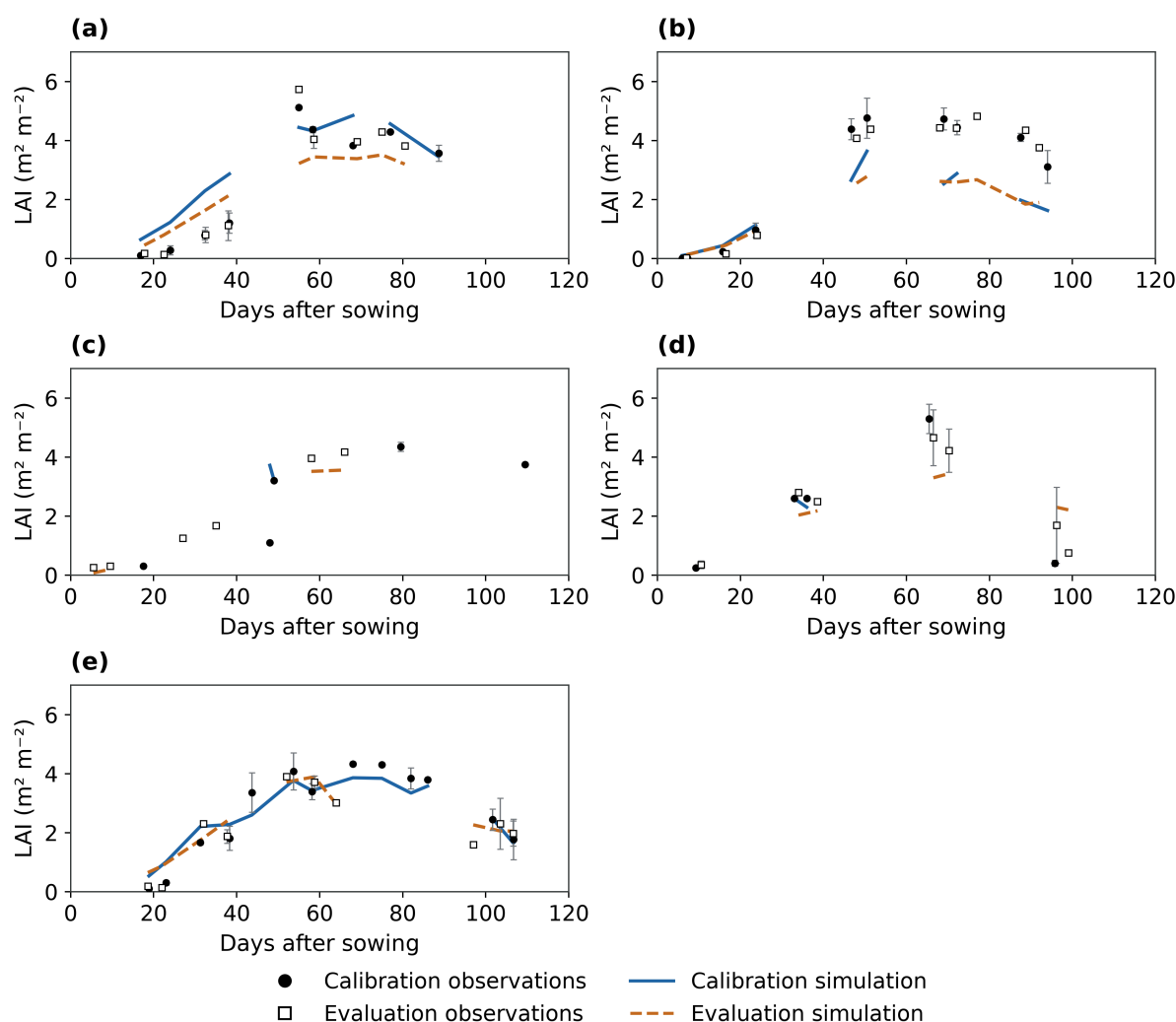
959 **Figure S6. Site-specific wheat LAI comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng. Calibration and**  
 960 **retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**  
 961 **errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



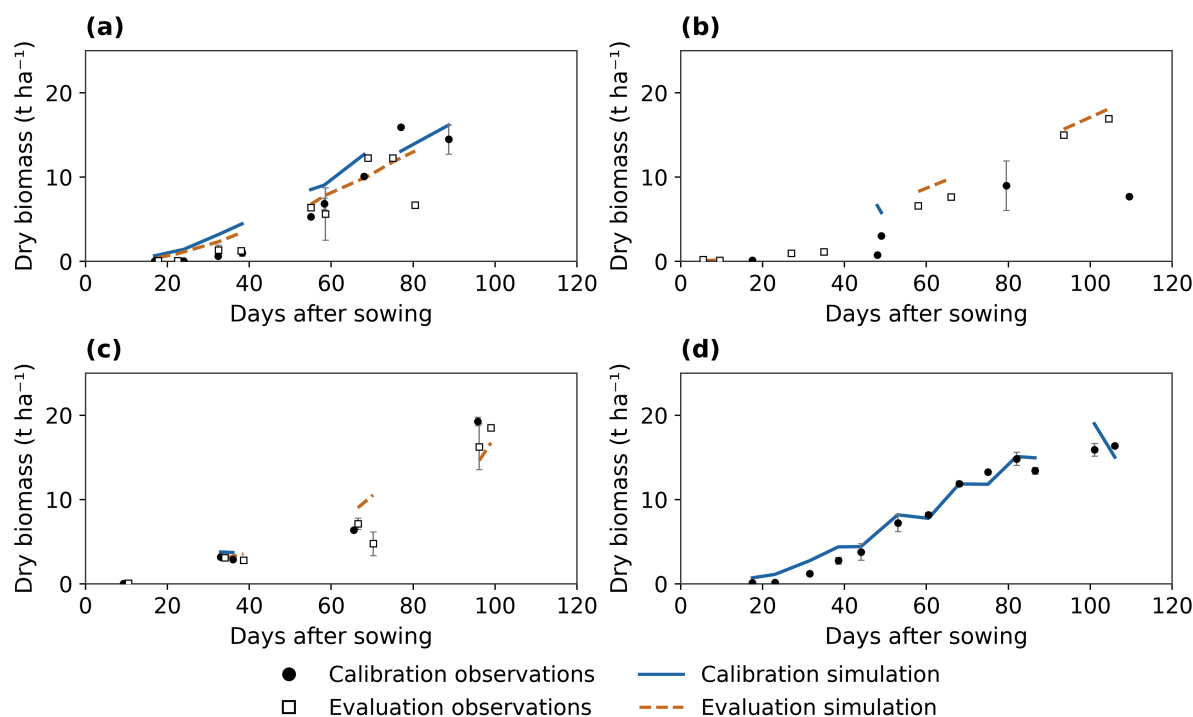
962 **Figure S7. Site-specific wheat aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**  
 963 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**  
 964 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



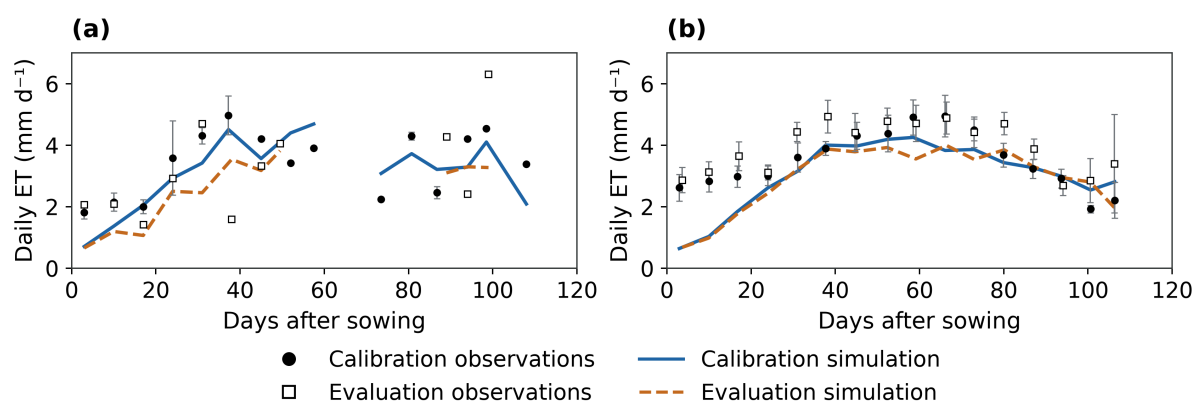
965 **Figure S8. Site-specific wheat ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**  
 966 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**  
 967 **weighting and conditional management interpretation follow Figure 4.**



968 **Figure S9. Site-specific maize LAI comparisons: (a) Fengqiu; (b) Gucheng; (c) Luancheng; (d) Shangqiu; (e) Yucheng. Calibration**  
 969 **and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-year**  
 970 **standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



971 **Figure S10. Site-specific maize aboveground biomass comparisons: (a) Fengqiu; (b) Luancheng; (c) Shangqiu; (d) Yucheng.**  
 972 **Calibration and retrospective-evaluation observations and simulations share the same axes within each site. Error bars describe site-**  
 973 **year standard errors; gaps are retained. The observations, weighting and conditional management interpretation follow Figure 4.**



974 **Figure S11. Site-specific maize ET comparisons: (a) Fengqiu; (b) Yucheng. Calibration and retrospective-evaluation observations and**  
 975 **simulations share the same axes within each site. Error bars describe site-year standard errors; gaps are retained. The observations,**  
 976 **weighting and conditional management interpretation follow Figure 4.**

## 977 **References**

978 Yang, X., Wang, G., Chen, Y., Sui, P., Pacenka, S., Steenhuis, T.S., Siddique, K.H.M., 2022. Reduced groundwater  
 979 use and increased grain production by optimized irrigation scheduling in winter wheat–summer maize double  
 980 cropping system—A 16-year field study in North China Plain. *Field Crops Res.* 275, 108364.

981 <https://doi.org/10.1016/j.fcr.2021.108364>

982 Yang, L., Fang, X., Zhou, J., Zhao, J., Hou, X., Yang, Y., Zang, H., Zeng, Z., 2024. Optimal irrigation for wheat-  
983 maize rotation depending on precipitation in the North China Plain: Evidence from a four-year experiment.  
984 *Agric. Water Manag.* 294, 108726. <https://doi.org/10.1016/j.agwat.2024.108726>

985 Zhang, X., Chen, S., Sun, H., Shao, L., Wang, Y., 2011. Changes in evapotranspiration over irrigated winter wheat  
986 and maize in North China Plain over three decades. *Agric. Water Manag.* 98, 1097–1104.  
987 <https://doi.org/10.1016/j.agwat.2011.02.003>

988 Wu, D., Fang, S., Li, X., He, D., Zhu, Y., Yang, Z., Xu, J., Wu, Y., 2019. Spatial-temporal variation in irrigation  
989 water requirement for the winter wheat-summer maize rotation system since the 1980s on the North China Plain.  
990 *Agric. Water Manag.* 214, 78–86. <https://doi.org/10.1016/j.agwat.2019.01.004>

991 Ren, P., Huang, F., Li, B., 2022. Spatiotemporal patterns of water consumption and irrigation requirements of  
992 wheat-maize in the Huang-Huai-Hai Plain, China and options of their reduction. *Agric. Water Manag.* 263,  
993 107468. <https://doi.org/10.1016/j.agwat.2022.107468>

994 Xiao, L., Wang, G., Wang, E., Liu, S., Chang, J., Zhang, P., Zhou, H., Wei, Y., Zhang, H., Zhu, Y., Shi, Z., Luo, Z.,  
995 2024. Spatiotemporal co-optimization of agricultural management practices towards climate-smart crop  
996 production. *Nature Food* 5, 59–71. <https://doi.org/10.1038/s43016-023-00891-x>

997 Jiang, Y., Xu, X., Huang, Q., Huo, Z., Huang, G., 2016. Optimizing regional irrigation water use by integrating a  
998 two-level optimization model and an agro-hydrological model. *Agricultural Water Management* 178, 76–88.  
999 <https://doi.org/10.1016/j.agwat.2016.08.035>

1000 Perry, C., Steduto, P., Allen, R.G., Burt, C.M., 2009. Increasing productivity in irrigated agriculture: Agronomic  
1001 constraints and hydrological realities. *Agric. Water Manag.* 96, 1517–1524.  
1002 <https://doi.org/10.1016/j.agwat.2009.05.005>

1003 Grafton, R.Q., Williams, J., Perry, C.J., Molle, F., Ringler, C., Steduto, P., Udall, B., Wheeler, S.A., Wang, Y.,  
1004 Garrick, D., Allen, R.G., 2018. The paradox of irrigation efficiency. *Science* 361, 748–750.  
1005 <https://doi.org/10.1126/science.aat9314>

1006 Chen, C., Wang, E., Yu, Q., 2010. Modelling the effects of climate variability and water management on crop water  
1007 productivity and water balance in the North China Plain. *Agric. Water Manag.* 97, 1175–1184.  
1008 <https://doi.org/10.1016/j.agwat.2008.11.012>

1009 Fang, Q.X., Ma, L., Flerchinger, G.N., Qi, Z., Ahuja, L.R., Xing, H.T., Li, J., Yu, Q., 2014. Modeling  
 1010 evapotranspiration and energy balance in a wheat–maize cropping system using the revised RZ-SHAW model.  
 1011 *Agricultural and Forest Meteorology* 194, 218–229. <https://doi.org/10.1016/j.agrformet.2014.04.009>

1012 Cammarano, D., Rötter, R.P., Asseng, S., Ewert, F., Wallach, D., Martre, P., Hatfield, J.L., Jones, J.W., Rosenzweig,  
 1013 C., Ruane, A.C., Boote, K.J., Thorburn, P.J., Kersebaum, K.C., Aggarwal, P.K., Angulo, C., Basso, B., Bertuzzi,  
 1014 P., Biernath, C., Brisson, N., Challinor, A.J., Doltra, J., Gayler, S., Goldberg, R., Heng, L., Hooker, J., Hunt,  
 1015 L.A., Ingwersen, J., Izaurralde, R.C., Müller, C., Kumar, S.N., Nendel, C., O’Leary, G.J., Olesen, J.E., Osborne,  
 1016 T.M., Palosuo, T., Priesack, E., Ripoche, D., Semenov, M.A., Shcherbak, I., Steduto, P., Stöckle, C.O.,  
 1017 Stratonovitch, P., Streck, T., Supit, I., Tao, F., Travasso, M., Waha, K., White, J.W., Wolf, J., 2016. Uncertainty  
 1018 of wheat water use: Simulated patterns and sensitivity to temperature and CO<sub>2</sub>. *Field Crops Research* 198, 80–  
 1019 92. <https://doi.org/10.1016/j.fcr.2016.08.015>

1020 Save, H., Bettadpur, S., Tapley, B.D., 2016. High-resolution CSR GRACE RL05 mascons. *Journal of Geophysical*  
 1021 *Research: Solid Earth* 121, 7547–7569. <https://doi.org/10.1002/2016jb013007>

1022 Scanlon, B.R., Zhang, Z., Save, H., Wiese, D.N., Landerer, F.W., Long, D., Longuevergne, L., Chen, J., 2016.  
 1023 Global evaluation of new GRACE mascon products for hydrologic applications. *Water Resources Research* 52,  
 1024 9412–9429. <https://doi.org/10.1002/2016wr019494>

1025 Long, D., Xu, Y., Cui, Y., Cui, Y., Butler, J.J., Dong, L., Wang, L., Liu, D., Wada, Y., Hu, L., Bai, G., Li, B., Wang,  
 1026 S., Nong, X., Cai, Y., Cheng, C., Mu, Y., Qiao, Y., Wang, J., Wang, H., Scanlon, B.R., 2025. Unprecedented  
 1027 large-scale aquifer recovery through human intervention. *Nat. Commun.* 16, 7296.  
 1028 <https://doi.org/10.1038/s41467-025-62719-5>

1029 Xu, Y., Long, D., Jiang, W., Cui, Y., Chen, F., Wang, J., Wada, Y., 2026. Quantifying Human and Climatic  
 1030 Contributions of Large-Scale Groundwater Recovery in the North China Plain. *Water Resour. Res.* 62,  
 1031 e2025WR043052. <https://doi.org/10.1029/2025wr043052>

1032 NASA JPL, 2013. NASA Shuttle Radar Topography Mission Global 1 arc second.  
 1033 <https://doi.org/10.5067/MEASURES/SRTM/SRTMGL1.003>

1034 Qiu, B., Hu, X., Chen, C., Tang, Z., Yang, P., Zhu, X., Yan, C., Jian, Z., 2022. Maps of cropping patterns in China  
 1035 during 2015–2021. *Scientific Data*. <https://doi.org/10.1038/s41597-022-01589-8>

1036 Natural Earth, n.d. Natural Earth, 1:10m Admin 1 – States, Provinces (version 5.1.1).

1037 Sun, H.-Y., Liu, C.-M., Zhang, X.-Y., Shen, Y.-J., Zhang, Y.-Q., 2006. Effects of irrigation on water balance, yield  
 1038 and WUE of winter wheat in the North China Plain. *Agric. Water Manag.* 85, 211–218.  
 1039 <https://doi.org/10.1016/j.agwat.2006.04.008>

1040 ISRIC — World Soil Information, 2020. SoilGrids250m version 2.0.

1041 Saxton, K.E., Rawls, W.J., 2006. Soil Water Characteristic Estimates by Texture and Organic Matter for Hydrologic  
 1042 Solutions. *Soil Science Society of America Journal* 70, 1569–1578. <https://doi.org/10.2136/sssaj2005.0117>

1043 Copernicus Climate Change Service, 2020. Agrometeorological indicators from 1979 to present derived from  
 1044 reanalysis. <https://doi.org/10.24381/CDS.6C68C9BB>

1045 Save, H., 2020. CSR GRACE and GRACE-FO RL06 Mascon Solutions v02. <https://doi.org/10.15781/cgq9-nh24>

1046 Long, D., Yang, W., Scanlon, B.R., Zhao, J., Liu, D., Burek, P., Pan, Y., You, L., Wada, Y., 2020. South-to-North  
 1047 Water Diversion stabilizing Beijing's groundwater levels. *Nat. Commun.* 11, 3665.  
 1048 <https://doi.org/10.1038/s41467-020-17428-6>

1049 Kiniry, J.R., Jones, C.A., O'toole, J.C., Blanchet, R., Cabelguenne, M., Spanel, D.A., 1989. Radiation-use efficiency  
 1050 in biomass accumulation prior to grain-filling for five grain-crop species. *Field Crops Research*.  
 1051 [https://doi.org/10.1016/0378-4290\(89\)90023-3](https://doi.org/10.1016/0378-4290(89)90023-3)

1052 de Wit; Berghuijs, n.d.. Python Crop Simulation Environment: WOFOST leaf dynamics implementation [software].  
 1053 Version Archived master-branch source snapshot.  
 1054 [https://github.com/ajwdewit/pcse/blob/master/pcse/crop/leaf\\_dynamics.py](https://github.com/ajwdewit/pcse/blob/master/pcse/crop/leaf_dynamics.py) (accessed 4 October 2026).

1055 Tan, M., Gou, F., Stomph, T.J., Wang, J., Yin, W., Zhang, L., Chai, Q., van der Werf, W., 2020. Dynamic process-  
 1056 based modelling of crop growth and competitive water extraction in relay strip intercropping: Model  
 1057 development and application to wheat-maize intercropping. *Field Crops Research* 246, 107613.  
 1058 <https://doi.org/10.1016/j.fcr.2019.107613>

1059 Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspiration: Guidelines for computing crop  
 1060 water requirements, FAO Irrigation and Drainage Paper. Food and Agriculture Organization of the United  
 1061 Nations, Rome.

1062 van Genuchten, M.T., 1980. A Closed-form Equation for Predicting the Hydraulic Conductivity of Unsaturated  
 1063 Soils. *Soil Science Society of America Journal* 44, 892–898.  
 1064 <https://doi.org/10.2136/sssaj1980.03615995004400050002x>

1065 Mualem, Y., 1976. A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water*  
 1066 *Resources Research* 12, 513–522. <https://doi.org/10.1029/wr012i003p00513>

1067 APSIM Initiative, 2026. APSIM Next Generation (ApsimX): wheat and maize model definitions [software].  
 1068 Version 074d15f.  
 1069 [https://github.com/APSIMInitiative/ApsimX/tree/074d15f51889a386d3667acfed184d03e4e8f65f/Models/Resou](https://github.com/APSIMInitiative/ApsimX/tree/074d15f51889a386d3667acfed184d03e4e8f65f/Models/Resources)  
 1070 *rces* (accessed 4 October 2026).

1071 Nash, J.E., Sutcliffe, J.V., 1970. River flow forecasting through conceptual models part I — A discussion of  
 1072 principles. *Journal of Hydrology* 10, 282–290. [https://doi.org/10.1016/0022-1694\(70\)90255-6](https://doi.org/10.1016/0022-1694(70)90255-6)

1073 Wu, L., Min, L., Liu, M., Zhang, Y., Pei, H., Li, H., Zhang, G., Wang, S., Shen, Y., 2023. Monitoring of Thick  
 1074 Vadose Zone Water Dynamics Under Irrigation Using a 48 m Deep Caisson at the Luancheng Critical Zone  
 1075 Observatory. *Water Resour. Res.* 59, e2022WR032965. <https://doi.org/10.1029/2022wr032965>

1076 Ines, A.V.M., Das, N.N., Hansen, J.W., Njoku, E.G., 2013. Assimilation of remotely sensed soil moisture and  
 1077 vegetation with a crop simulation model for maize yield prediction. *Remote Sens. Environ.* 138, 149–164.  
 1078 <https://doi.org/10.1016/j.rse.2013.07.018>

1079 Roy, A., Narvekar, P., Murtugudde, R., Shinde, V., Ghosh, S., 2021. Short and Medium Range Irrigation Scheduling  
 1080 Using Stochastic Simulation-Optimization Framework With Farm-Scale Ecohydrological Model and Weather  
 1081 Forecasts. *Water Resources Research* 57, e2020WR029004. <https://doi.org/10.1029/2020wr029004>

1082 Meier, U. (Ed.), 2018. Growth stages of mono- and dicotyledonous plants: BBCH Monograph. Julius Kühn-Institut,  
 1083 Quedlinburg. <https://doi.org/10.5073/20180906-074619>

1084 de Wit, A., 2020. Parameter sets for the WOFOST 7.2/7.3 cropping system model for wheat.

1085 de Wit, A., 2022. Parameter sets for the WOFOST 7.2/7.3 model for simulating maize.

1086 Fu, Y., Dong, J., Shen, R., Peng, Q., Li, X., Zheng, Y., Pan, B., Dong, J., Qiu, B., Huang, J., Yuan, W., 2025. China  
 1087 Crop Dataset: A long-term and high-resolution crop distribution dataset in China.  
 1088 <https://doi.org/10.57760/sciencedb.32361>

1089 Fu, Y., Dong, J., Shen, R., Peng, Q., Li, X., Zheng, Y., Pan, B., Dong, J., Qiu, B., Huang, J., Huang, X., Yuan, W.,  
 1090 2026. A 30-m annual distribution dataset of major crops in China from 2001-2024. *Scientific Data* 13, 1074.  
 1091 <https://doi.org/10.1038/s41597-026-07370-5>